

O-RAN White Box Hardware Working Group Indoor Picocell Hardware Architecture and Requirement (FR1 Only) Specification

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1

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2

Contents

1		
2	Revision History	2
3	Chapter 1 Introductory Material	7
4	1.1 Scope	7
5	1.2 References	7
6	1.3 Definitions and Abbreviations	8
7	1.3.1 Definitions.....	8
8	1.3.2 Abbreviations.....	8
9	Chapter 2 Deployment Scenarios and White Box Base Station Architecture	11
10	2.1 Deployment Scenarios	11
11	2.2 White Box Base Station Architecture	11
12	2.2.1 Split RAN Architecture.....	12
13	2.2.2 Integrated gNB-DU Architecture	14
14	2.2.3 All-In-One Base Station Architecture	14
15	Chapter 3 White Box Hardware Architecture	15
16	3.1 O-CU Hardware Architecture.....	15
17	3.1.1 O-CU Architecture Diagram	15
18	3.1.2 O-CU Functional Module Description	15
19	3.1.3 O-CU Interfaces	15
20	3.2 O-DU ₇₋₂ Hardware Architecture	15
21	3.2.1 O-DU ₇₋₂ Architecture Diagram	16
22	3.2.2 O-DU ₇₋₂ Functional Module Description	16
23	3.2.3 O-DU ₇₋₂ Interfaces	17
24	3.3 O-DU ₆ Hardware Architecture	18
25	3.3.1 O-DU ₆ Architecture Diagram.....	18
26	3.3.2 O-DU ₆ Functional Module Description	18
27	3.3.3 O-DU ₆ Interfaces.....	18
28	3.4 O-DU ₈ Hardware Architecture.....	18
29	3.4.1 O-DU ₈ Architecture Diagram.....	19
30	3.4.2 O-DU ₈ Functional Module Description	19
31	3.4.3 O-DU ₈ Interfaces	20
32	3.5 FHGW ₇₋₂ - Option 7-2 to Option 7-2 Hardware Architecture.....	20
33	3.5.1 FHGW ₇₋₂ Architecture Diagram.....	20
34	3.5.2 FHGW ₇₋₂ Functional Module Description.....	20
35	3.6 FHGW ₆ - Option 6 to Option 6 Hardware Architecture	21
36	3.7 FHGW _{7-2->8} - Option 7-2 to Option 8 Hardware Architecture.....	21
37	3.7.1 FHGW _{7-2->8} Architecture Diagram	21
38	3.7.2 FHGW _{7-2->8} Functional Module Description.....	22
39	3.8 FHGW ₈ - Option 8 to Option 8 Hardware Architecture	22
40	3.8.1 FHGW ₈ Architecture Diagram.....	22
41	3.8.2 FHGW ₈ Functional Module Description	23
42	3.9 O-RU ₇₋₂ Hardware Architecture.....	24
43	3.9.1 O-RU ₇₋₂ Architecture Diagram.....	24
44	3.9.2 O-RU ₇₋₂ Functional Module Description.....	24
45	3.10 O-RU ₆ Hardware Architecture.....	25
46	3.10.1 O-RU ₆ Architecture Diagram.....	25
47	3.10.2 O-RU ₆ Functional Module Description.....	25
48	3.11 O-RU ₈ Hardware Architecture	26

1	3.11.1	O-RU ₈ Architecture Diagram	26
2	3.11.2	O-RU ₈ Functional Module Description.....	27
3	3.12	Integrated gNB-DU Hardware Architecture	27
4	3.12.1	Integrated gNB-DU Architecture Diagram	28
5	3.12.2	Integrated gNB-DU Function Module Description	28
6	3.13	AIO White Box Hardware Architecture	29
7	3.13.1	AIO Architecture Diagram.....	29
8	3.13.2	AIO Functional Module Description.....	29
9	3.13.3	AIO Interfaces.....	30
10	Chapter 4	White Box Hardware Requirements	31
11	4.1	O-CU Requirements	31
12	4.1.1	O-CU Performance.....	31
13	4.1.2	O-CU Interfaces	32
14	4.1.3	O-CU Environmental and EMC.....	32
15	4.1.4	O-CU Mechanical, Thermal and Power.....	32
16	4.2	O-DU _x Common Requirements	33
17	4.2.1	O-DU _x Performance	34
18	4.2.2	O-DU _x Interfaces.....	35
19	4.2.3	O-DU _x Environmental and EMC	35
20	4.2.4	O-DU _x Mechanical, Thermal and Power	35
21	4.3	O-DU _x Split Option Specific Requirements	36
22	4.3.1	O-DU ₇₋₂ Specific Requirements	36
23	4.3.2	O-DU ₆ Specific Requirements	37
24	4.3.3	O-DU ₈ Specific Requirements	37
25	4.4	O-RU _x Common Requirements	37
26	4.4.1	O-RU _x Performance	37
27	4.4.2	O-RU _x Interfaces.....	39
28	4.4.3	O-RU _x Environmental and EMC.....	39
29	4.4.4	O-RU _x Mechanical, Thermal and Power.....	39
30	4.5	O-RU _x Split Option Specific Requirements.....	40
31	4.5.1	O-RU ₇₋₂ Specific Requirements	40
32	4.5.2	O-RU ₆ Specific Requirements	40
33	4.5.3	O-RU ₈ Specific Requirements	41
34	4.6	FHGW _x – Common Requirements	41
35	4.6.1	FHGW _x Performance	41
36	4.6.2	FHGW _x Interfaces.....	41
37	4.6.3	FHGW _x Environmental, EMCs.....	42
38	4.6.4	FHGW _x Mechanical, Thermal and Power.....	42
39	4.7	FHGW _x – Split Option Specific Requirements.....	43
40	4.7.1	FHGW ₇₋₂ Specific Requirements.....	43
41	4.7.2	FHGW _{7-2->8} Specific Requirements.....	43
42	4.7.3	FHGW ₈ Specific Requirements.....	43
43	4.8	Integrated gNB-DU Requirements	43
44	4.8.1	Integrated gNB-DU Performance.....	44
45	4.8.2	Integrated gNB-DU Interfaces	45
46	4.8.3	Integrated gNB-DU Environmental, EMC.....	45
47	4.8.4	Integrated gNB-DU Mechanical, Thermal and Power	46
48	4.9	AIO White Box Hardware Requirements	47
49	4.9.1	AIO Performance	47
50	4.9.2	AIO Interfaces.....	48
51	4.9.3	AIO Environmental and EMC.....	49

4.9.4	AIO Mechanical, Thermal and Power	49
-------	---	----

Tables

Table 4-1	:O-CU Performance Requirements	31
Table 4-2	:O-CU Interface Requirements	32
Table 4-3	:O-CU EMC Requirements	32
Table 4-4	:O-CU Mechanical Requirements	32
Table 4-5	:O-CU Thermal Requirements	33
Table 4-6	:O-CU Power Requirements	33
Table 4-7	:O-DU _x Performance Requirements	34
Table 4-8	:O-DU _x Interface Requirements	35
Table 4-9	:O-DU _x EMC Requirements	35
Table 4-10	:O-DU _x Mechanical Requirements	35
Table 4-11	:O-DU _x Thermal Requirements	36
Table 4-12	:O-DU _x Power Requirements	36
Table 4-13	:O-DU ₇₋₂ Specific Requirements	36
Table 4-14	:O-DU ₆ Specific Requirements	37
Table 4-15	:O-DU ₈ Specific Requirements	37
Table 4-16	:O-RU _x Performance Requirements	38
Table 4-17	:O-RU _x Interface Requirements	39
Table 4-18	:O-RU _x Environmental and EMC Requirements	39
Table 4-19	:O-RU _x Mechanical, Thermal and Power Requirements	39
Table 4-20	:FHGW _x Performance Requirements	41
Table 4-21	:FHGW _x Interface Requirements	41
Table 4-22	:FHGW _x EMC Requirements	42
Table 4-23	:FHGW _x Mechanical, Thermal and Power Requirements	42
Table 4-24	:Integrated gNB-DU Performance Requirements	44
Table 4-25	:Integrated gNB-DU Interface Requirements	45
Table 4-26	:Integrated gNB-DU Environmental and EMC Requirements	45
Table 4-27	:Integrated gNB-DU Mechanical, Thermal and Power Requirements	46
Table 4-28	:AIO Whitebox Performance Requirements	47
Table 4-29	:AIO Whitebox Interface Requirements	48
Table 4-30	:AIO EMC Requirements	49
Table 4-31	:AIO Mechanical Requirements	49
Table 4-32	:Thermal Requirements	49
Table 4-33	:AIO Whitebox Power Requirements	50

Figures

Figure 2-1	:Indoor Split Architecture	11
Figure 2-2	:Indoor Integrated Architecture	12
Figure 2-3	:Indoor All-In-One Architecture	12
Figure 2-4	:Option 6 to Option 6 Split Architecture	12
Figure 2-5	:Option 7-2 to Option 7-2 Split Architecture	13

1	Figure 2-6: Option 7-2 to Option 8 Split Architecture	13
2	Figure 2-7: Option 8 to Option 8 Split Architecture	14
3	Figure 2-8: Integrated gNB-DU Architecture	14
4	Figure 2-9: All-In-One Base Station Architecture	14
5	Figure 3-1: O-CU White Box Hardware Block Diagram.....	15
6	Figure 3-2: O-DU ₇₋₂ with Split Physical Function.....	16
7	Figure 3-3: O-DU ₇₋₂ Functional Block and Interface Diagram.....	17
8	Figure 3-4: O-DU ₆ Architecture Diagram	18
9	Figure 3-5: O-DU ₈ Architecture Diagram	19
10	Figure 3-6: O-DU ₈ Functional Block and Interface Diagram	19
11	Figure 3-7: FHGW ₇₋₂ Architecture Diagram	20
12	Figure 3-8: FHGW ₇₋₂ Functional Module Diagram.....	21
13	Figure 3-9: FHGW _{7-2->8} Architecture Diagram.....	21
14	Figure 3-10: FHGW _{7-2->8} Functional Module Diagram	22
15	Figure 3-11: FHGW ₈ Architecture Diagram	23
16	Figure 3-12: FHGW ₈ Functional Module Diagram	23
17	Figure 3-13: O-RU ₇₋₂ Architecture Diagram	24
18	Figure 3-14: O-RU ₇₋₂ Functional Module Diagram.....	25
19	Figure 3-15: O-RU ₆ Architecture Diagram	25
20	Figure 3-16: O-RU ₆ Functional Module Diagram.....	26
21	Figure 3-17: O-RU ₈ Architecture Diagram.	27
22	Figure 3-18: O-RU ₈ Functional Module Diagram	27
23	Figure 3-19: Integrated gNB-DU Architecture Diagram.....	28
24	Figure 3-20: Integrated gNB-DU Functional Module Diagram.....	28
25	Figure 3-21: AIO White box Architecture Diagram	29
26		

Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

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- x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).
- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the Indoor Picocell deployment scenario case as specified in the Deployment Scenarios and Base Station Classes document [1].

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- [1] ORAN.WG7.DSC.0-v05.00 Technical Specification, ‘Deployment Scenarios and Base Station Classes for White Box Hardware’.
- [2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [3] 3GPP TR 38.104: "NR; Base Station (BS) radio transmission and reception".
- [4] ORAN-WG4.CUS.0-v01.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’.
- [5] CPRI Specification V7.0 (2015-10-09) Interface Specification, ‘Common Public Radio Interface (CPRI).
- [6] Small Cell Forum nFAPI Specification (Not yet publicly available, however it will be available as part of an O-RAN approved WG7 reference design specification)
- [7] 3GPP TS 38.113: “NR: Base Station (BS) Electromagnetic Compatibility (EMC)
- [8] O-RAN.SuFG.CE-v01.00, O-RAN SuFG, “Circular economy guidelines on network equipment – Technical Report”, - v01.00.

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [2] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

All-In-One architecture: In the all-in-one architecture, the O-RU, O-DU and O-CU are implemented on one platform. There is no need for neither fronthaul interface between O-RU and O-DU nor F1 interface between O-DU and O-CU.

Carrier Frequency: Center frequency of the cell.

F1 interface: The open interface between O-CU and O-DU_x.

Integrated architecture: In the integrated architecture, the O-RU_x and O-DU_x are implemented on one platform. Each O-RU_x and RF front end is associated with one O-DU_x. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU_x and O-DU_x are physically separated from one another in this architecture. A switch may aggregate multiple O-RU_x (s) to one O-DU_x. O-DU_x, switch and O-RU_x (s) are connected by the fronthaul interface as defined in WG4.

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
5GC	5G Core
ACS	Adjacent Channel Selectivity
ADC	Analog to Digital Converter
AIO	All-In-One
ASIC	Application Specific Integrated Circuit
ATA	Advanced Technology Attachment
BB	Baseband
BPSK	Binary Phase Shift Keying
BS	Base Station
CFR	Crest Factor Reduction
CU	Centralized Unit as defined by 3GPP
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DL	Downlink

1	DPD	Digital Pre-Distortion
2	DSP	Digital Signal Processor
3	DU	Distributed Unit as defined by 3GPP
4	DUC	Digital Up Conversion
5	EMC	ElectroMagnetic Compatibility
6	EVM	Error Vector Magnitude
7	FFT	Fast Fourier Transform
8	FH	Fronthaul
9	FHGW	Fronthaul Gateway
10	FHGW _x	Fronthaul gateway with no FH protocol translation, supporting an O-DU _x with split option x and an O-RU _x with split option x, with currently available options 6→6, 7-2→7-2 and 8→8
13	FHGW _{x→y}	Fronthaul Gateway that can translate FH protocol from an O-DU _x with split option x to an O-RU _y with split option y, with currently available option 7-2→8.
15	FPGA	Field Programmable Gate Array
16	GbE	Gigabit Ethernet
17	GPP	General Purpose Processor
18	IEEE	Institute of Electrical and Electronics Engineers
19	IMD	InterModulation Distortion
20	I/O	Input/Output
21	JTAG	Joint Test Action Group
22	L1	Layer 1, also referred as PHY, also known as Physical Layer of Open System Interconnection (OSI) model
24	L2	Layer 2, also referred to as Data Link layer in OSI model
25	L3	Layer 3, also referred to as Network Layer in OSI model
26	LED	Light Emitting Diode
27	LTE	Long Term Evolution
28	MAC	Media Access Control
29	MIMO	Multiple Input Multiple Output
30	MCP	Multi-Chip Package
31	MU-MIMO	Multiple User MIMO
32	NG	Next Generation
33	NR	New Radio
34	OAM	Operations, Administrations and Maintenance
35	O-CU	O-RAN Centralized Unit as defined by O-RAN
36	O-DU _x	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2 (as defined by WG4) or 8
38	O-RU _x	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2 (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DU _x
41	PCIe	Peripheral Component Interface Express
42	PDCCP	Packet Data Convergence Protocol
43	PHY	Physical Layer, also referred as L1
44	PLL	Phase Locked Loop
45	POE	Power over Ethernet
46	QAM	Quadrature Amplitude Modulation
47	QPSK	Quadrature Phase Shift Keying
48	RAN	Radio Access Network

1	RF	Radio Frequency
2	RLC	Radio Link Controller
3	RRC	Radio Resource Controller
4	RU	Radio Unit as defined by 3GPP
5	RX	Receiver
6	SDU	Service Data Unit
7	SFP	Small Form-factor Pluggable
8	SFP+	Small Form-factor Pluggable Transceiver
9	SoC	System on Chip
10	SPI	Serial Peripheral Interface
11	TR	Technical Report
12	TS	Technical Specification
13	TX	Transmitter
14	UL	Uplink
15	USB	Universal Serial Bus
16	WG	Working Group

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Chapter 2 Deployment Scenarios and White Box Base Station Architecture

This chapter consists of two parts: the deployment scenario and the white box architecture. The deployment scenarios outline more specific functional requirements of the base station. All the reference designs shall meet these requirements in order to comply with O-RAN white box standard. In the white box hardware architecture section, it describes the overall gNB hardware architecture and function partition that meet the design requirements. The details on each of these topics are described in the following sections.

2.1 Deployment Scenarios

The indoor picocell is the targeted deployment scenario of this specification. The requirements of the indoor picocell base station are listed in the white box Deployment Scenarios and Base Station Classes document [1]. Some of the key requirements described in that document are highlighted here.

- Cell type: Indoor
- Carrier Frequency Band: FR1
- Frequency Bandwidth: Up to 100 MHz
- Inter site distance: 20 meters
- Antennas: 2Tx2Rx; 4Tx4Rx
- Fronthaul Type: O-RAN FH (WG4), 3GPP Option 6 and 3GPP Option 8

2.2 White Box Base Station Architecture

In general, the base station hardware architecture can be classified by using different criteria. The physical partition method is adopted by O-RAN; hence, the base station architecture is divided into three categories namely split, integrated, and all-in-one. In split architecture, the fronthaul interface determines the gNB functions location. Here we refer to the two partitions as O-DU_x and O-RU_x; where the “x” is split option number. In case the O-RAN WG4 defined fronthaul interface is used, these two partitions are called O-DU₇₋₂ and O-RU₇₋₂. For a more complete description of the terminology used, refer to the Deployment Scenarios and Base Station Classes Document [1].

The split architecture is shown in Figure 2-1. A Fronthaul Gateway (FHGW) is an optional device between the O-DU_x and O-RU_x to aggregate multiple radio units together. Within this specification, a Back End fronthaul interface is defined as the connection between the FHGW_x and the O-DU_x, while the Font End fronthaul interface is defined as the connection between the FHGW_x and the O-RU_x. With split architectures, one may choose to have O-CU and O-DU_x as either co-located or shared configuration with respect to hardware.

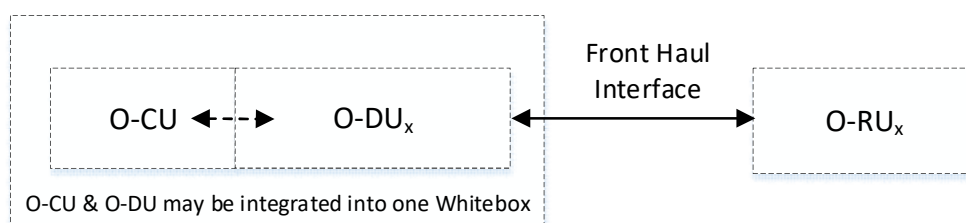


Figure 2-1: Indoor Split Architecture

For integrated base station architecture, the O-DU is integrated with the O-RU into one box. Figure 2-2 shows the integrated architecture.

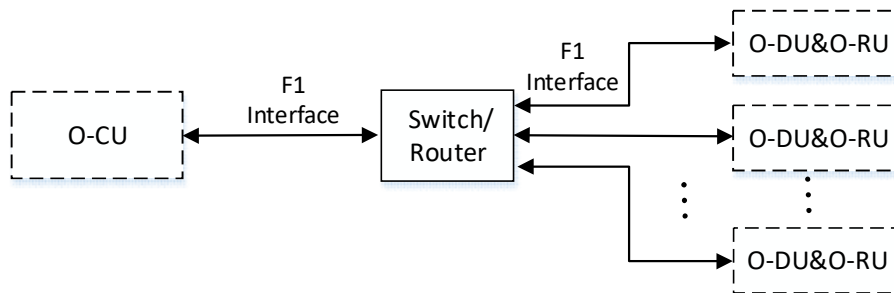


Figure 2-2: Indoor Integrated Architecture

The all-in-one architecture is shown in Figure 2-3. This architecture is defined as where O-CU, O-DU_x and O-RU_x are physically located in one hardware box.

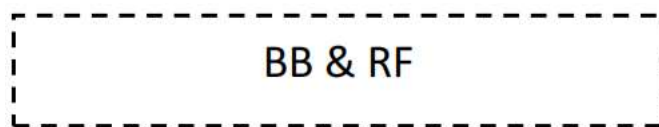


Figure 2-3: Indoor All-In-One Architecture

2.2.1 Split RAN Architecture

For medium and large coverage indoor deployment scenarios, the base station with split RAN architecture is more cost effective. This type of architecture is widely deployed in 4G networks. The main idea of this architecture is “shared cell”. By using the fronthaul gateway, a large number of radio units can then share the same radio spectrum resource in one cell. This configuration is very useful in both low and high capacity scenarios. Therefore, when an O-DU_x and FHGW_x are capable of handling cell splits, multiple O-RU_x(s) can be grouped together to share the radio resources. Accordingly, the choice of fronthaul split option changes the architecture of the base station. WG7 has recognized the following four split option architectures per service providers’ deployment needs which are shown in Figure 2-4 through Figure 2-7.

1) Option 6 to option 6 split architecture

In split option 6, the L1 functions are within O-RU₆ while the O-DU₆ contains functions of MAC and above. Figure 2-4 shows the block diagram of base station having an option 6 split architecture.

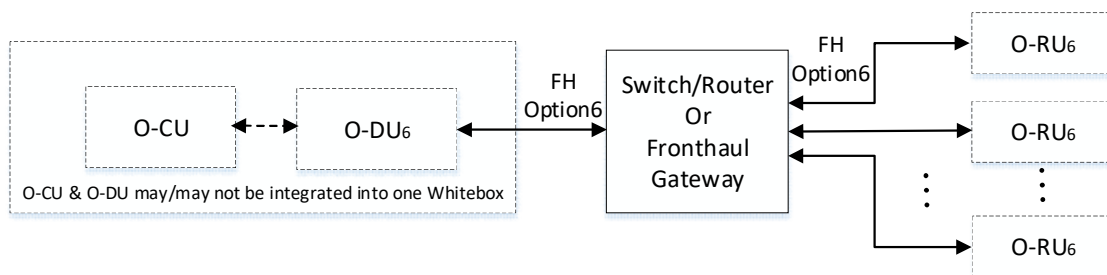


Figure 2-4: Option 6 to Option 6 Split Architecture

2) Option 7-2 to option 7-2 split architecture:

In split option 7-2, low PHY functions reside in the O-RU₇₋₂, while the high PHY functions reside in the O-DU₇₋₂. O-RAN WG4 CUS-plane spec [4] outlines the details of this split option under different usage scenarios.

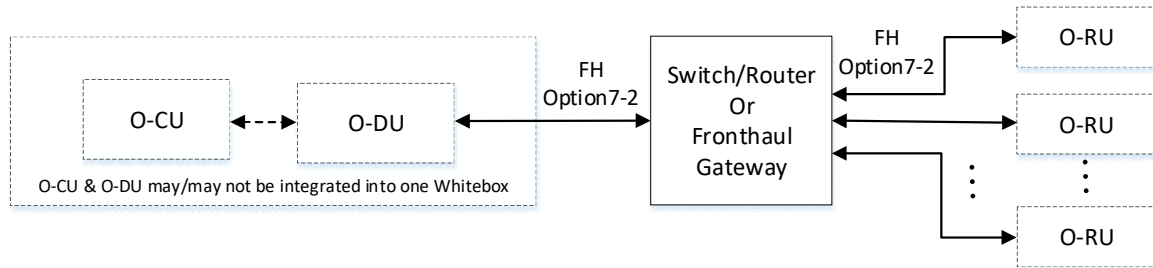


Figure 2-5: Option 7-2 to Option 7-2 Split Architecture

3) Option 7-2 to option 8 split:

This configuration is selected when a deployment scenario requires a radio unit which only supports split option 8 architecture. Since this is currently not part of overall O-RAN architecture, CPRI based option 8 fronthaul and FHGW_{7-2→8} will be included in the reference design specification. Figure 2-6 depicts the option 7-2 to option 8 based split architecture. There is no change for the definition of O-DU₇₋₂, However, O-RU₈ supports option 8 fronthaul interface while the FHGW_{7-2→8} translates the fronthaul protocol between option 7-2 and CPRI.

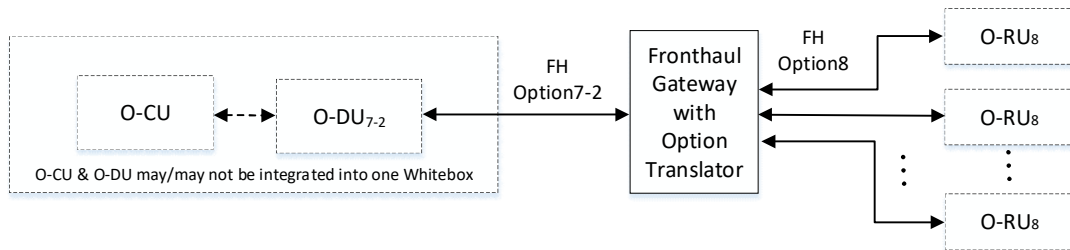


Figure 2-6: Option 7-2 to Option 8 Split Architecture

4) Option 8 to option 8 split architecture:

Currently, Option 8 is a non-O-RAN defined split option where the CPRI fronthaul interface is needed in order to make the interoperability work with O-DU₈ and O-RU₈ from different vendors. The option 8 fronthaul interface definition and the requirements shall be part of the white box reference design, if adopted. In this case, the O-DU₈ consists of L1 and L2 processing functions. Figure 2-7 shows an option 8 based split architecture. Note that O-CU and O-DU₈ hardware may be integrated into one Whitebox this is also shown in Figure 2-7.

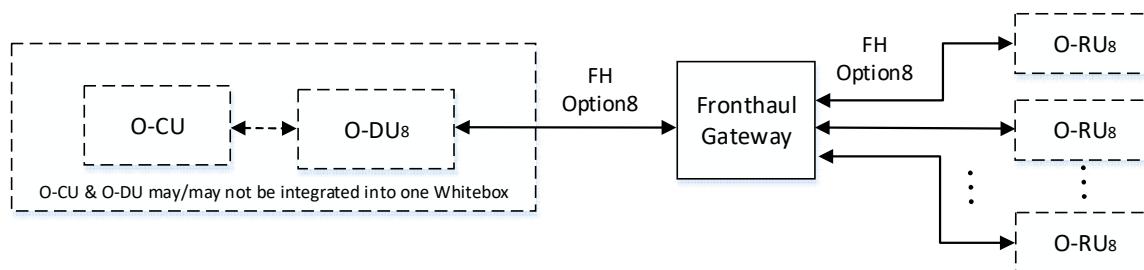


Figure 2-7: Option 8 to Option 8 Split Architecture

2.2.2 Integrated gNB-DU Architecture

For integrated gNB-DU, the complete L1/L2 and radio functions are integrated into a single box which includes all logical functions of O-DU and O-RU. 错误!未找到引用源。 shows the block diagram of integrated gNB-DU. The gNB-DU connects with the O-CU through an F1 interface as defined by 3GPP.

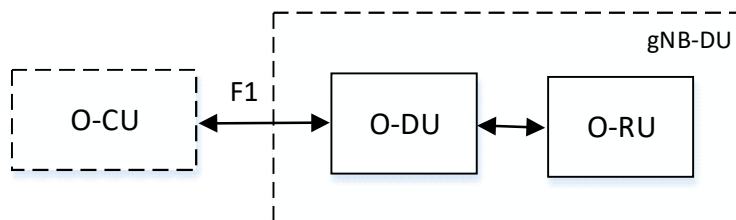


Figure 2-8: Integrated gNB-DU Architecture

2.2.3 All-In-One Base Station Architecture

For all-in-one base station, the functions of O-CU, O-DU_x and O-RU_x are included in one box which is composed of baseband unit and RF processing unit, as such no FH interface and F1 interface are required to connect them.

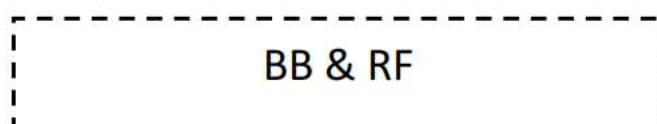


Figure 2-9: All-In-One Base Station Architecture

Chapter 3 White Box Hardware Architecture

Based on the gNB physical implementation architectures discussed earlier, this chapter provides the architecture, major building blocks and all external/internal interfaces for each Whitebox.

3.1 O-CU Hardware Architecture

In 3GPP system architecture, the gNB Central Unit (CU) communicates to the Distribution Unit (DU) via an F1 interface. This interface has been adopted by O-RAN Alliance as well. F1 is an IP based protocol interface, which offers more flexibility on O-CU platform design.

3.1.1 O-CU Architecture Diagram

The O-CU can be implemented with any General-Purpose Processor (GPP) based platform having an optional accelerator block. The O-CU functions can be implemented in either a separated hardware platform or share the same hardware platform with O-DU_x functions (Integrated). In both cases, the O-CU should be able to leverage the O-DU_x reference design with relaxed requirements on I/O and network bandwidth capacity. Figure 3-1 shows the hardware blocks and interfaces within the O-CU white box. Refers to the O-DU_x section for the details of O-CU components and interfaces.

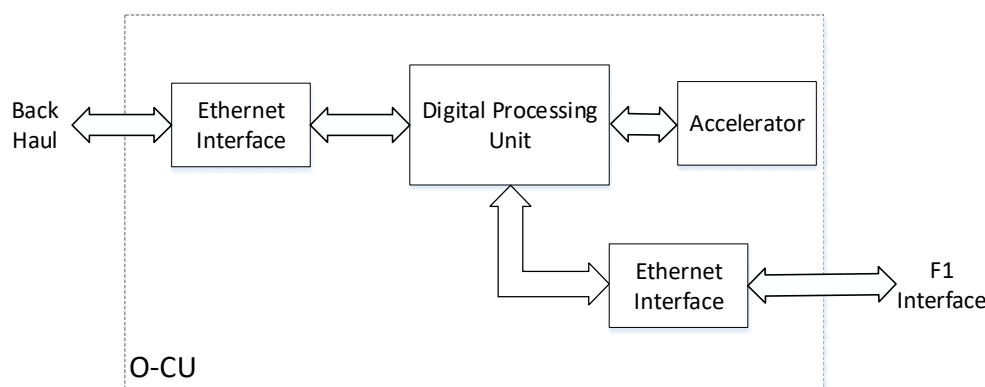


Figure 3-1: O-CU White Box Hardware Block Diagram

3.1.2 O-CU Functional Module Description

The O-CU functional module description is the same as for the O-DU₇₋₂ and it is described in Section 3.2.2.

3.1.3 O-CU Interfaces

The O-CU interfaces are the same as for the O-DU₇₋₂ and it is described in Section 3.2.3.

3.2 O-DU₇₋₂ Hardware Architecture

For split RAN architecture, the functional blocks of RAN physical layer are divided into two parts – high PHY and low PHY. The O-RAN lower layer split is defined in O-RAN WG4 fronthaul interface specification [4]. It also provides the details of the interface protocol as well as function partitions.

3.2.1 O-DU₇₋₂ Architecture Diagram

Depending on the 3GPP standards and category of the radio unit, the split function blocks within O-DU₇₋₂ and O-RU₇₋₂ may vary accordingly. The O-RAN fronthaul C/U/S-plane specification [4] offers comprehensive information on this topic. The hardware functional partition architecture is shown in Figure 3-2.

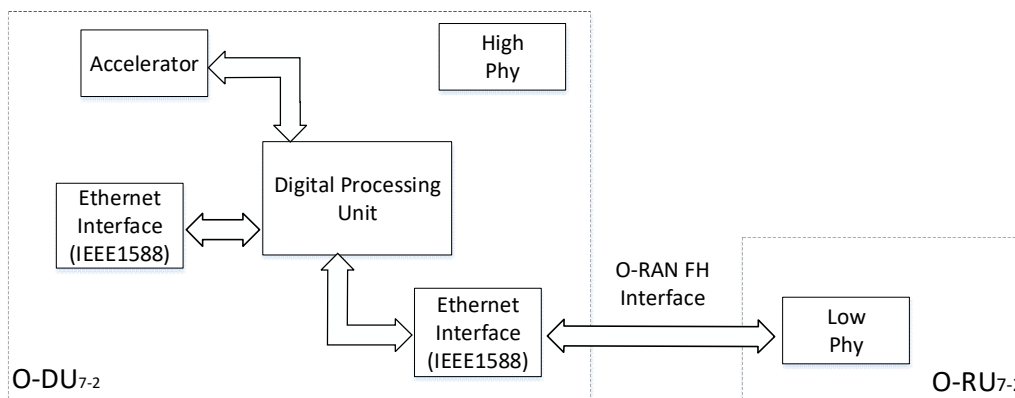


Figure 3-2: O-DU₇₋₂ with Split Physical Function

3.2.2 O-DU₇₋₂ Functional Module Description

O-DU₇₋₂ or O-CU hardware components selection is a product design specific task. Figure 3-3 shows the required components of O-DU₇₋₂. Their descriptions and requirements are as follows:

Digital Processing Unit

The processing unit can be any GPP or digital signal processor (DSP), with I/O chipset. It may also be in the form of System-On-Chip (SOC), or Multi-Chip Package (MCP).

Memory

DDR memory devices are used to store the runtime data and software for the processing unit.

Flash Memory

On board non-volatile storage device is used to store the firmware and non-volatile data, such as log data.

Board Management Controller

The controller is used to manage/control the power and monitors the operational status of the board.

Storage Device

The storage device such as hard drive is used to store OS, driver and applications software.

Ethernet Controller

The Ethernet ports transport the fronthaul or backhaul traffic according to the gNB hardware node requirements. The Ethernet device shall support IEEE1588 based timing synchronization.

Accelerator

The accelerator is an optional device. For performance improvement, hardware accelerator can be used for any process function (e.g. Forward Error Correction (FEC)).

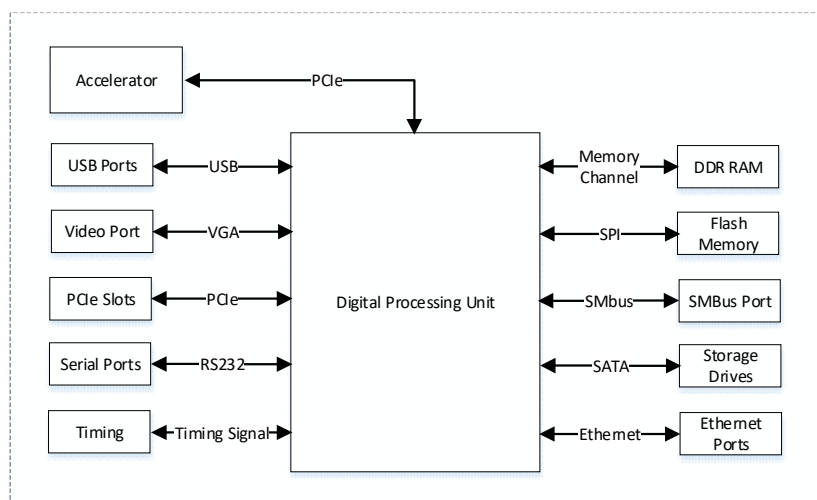


Figure 3-3: O-DU₇₋₂ Functional Block and Interface Diagram.

3.2.3 O-DU₇₋₂ Interfaces

The O-DU₇₋₂ supported interfaces described below are also shown in Figure 3-3.

Memory Channel Interface

Support DDR4 and later memory interface.

PCIe Interface

Support for PCIe v3 and later interface; the bandwidth depends on the use cases, and it can be used to connect an accelerator device or network card.

Ethernet Interfaces

Supports any one or combination of GbE/10GbE/25GbE/40GbE links.

Serial ATA Interface

SATA3 shall be supported in case of software storage, such as hard drive.

SPI Interface

The SPI interface connects the processor with flash type of device for firmware.

Video Interface

Video interface is optional.

USB Interface

Used to connect with local device for debug or on-board firmware update.

Miscellaneous Interface

Other interfaces that may be needed such as serial port, JTAG, etc.

3.3 O-DU₆ Hardware Architecture

Hardware architecture for base stations deploying architecture with split option 6 are described in the following sections.

3.3.1 O-DU₆ Architecture Diagram

O-DU₆ and O-RU₆ are connected via Ethernet based nFAPI interface[6]. O-RU₆ includes complete physical layer processing functions, thus the O-DU₆ handles L2 and L3 processing functions. The O-DU₆ hardware architecture is shown in Figure 3-4.

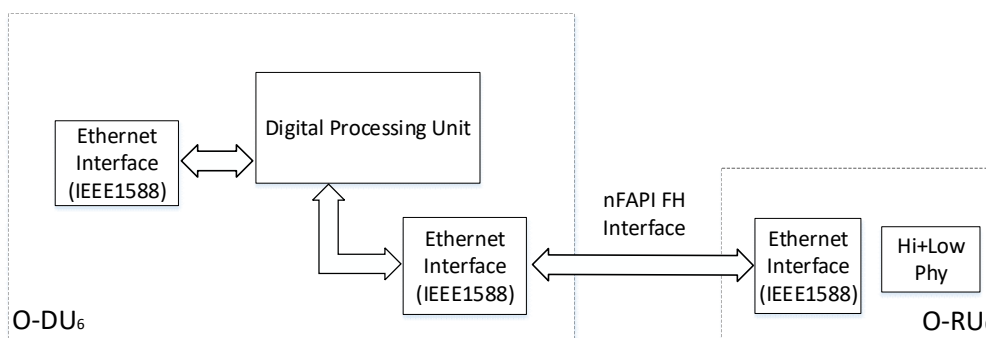


Figure 3-4: O-DU₆ Architecture Diagram

3.3.2 O-DU₆ Functional Module Description

O-DU₆ hardware functional blocks are the same as the modules used by O-DU₇₋₂. A detailed description of O-DU₇₋₂ modules is given in Section 3.2.2. However, note that O-DU₆ does not require an accelerator module since all physical layer functions are handled by O-RU₆.

3.3.3 O-DU₆ Interfaces

The O-DU₆ interfaces are the same as O-DU₇₋₂ interfaces which are described in section 3.2.3.

3.4 O-DU₈ Hardware Architecture

Base stations deploying split architecture option 8 have their baseband processing and radio processing units physically separated from one another. For this type of functional partition, CPRI must be used as fronthaul interface between O-DU₈ and O-RU₈. The architecture for O-DU₈ is described in the following sections

3.4.1 O-DU₈ Architecture Diagram

O-DU₈ shall contain complete physical layer processing operation. The details of CPRI interface will be provided as part of the reference design if this option is selected by the system designer. The hardware functional partition architecture is shown in Figure 3-5.

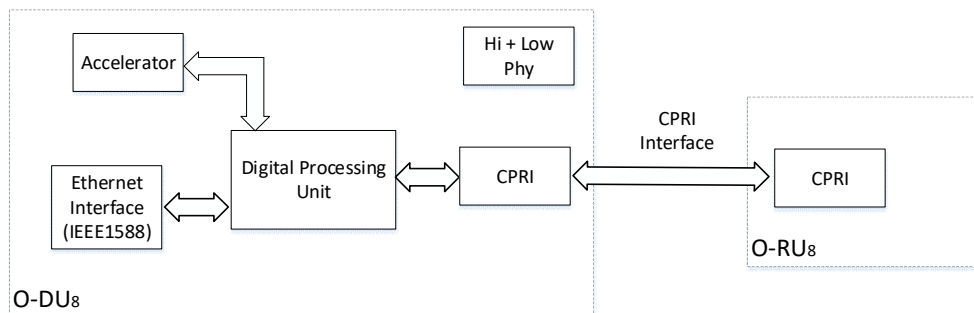


Figure 3-5: O-DU₈ Architecture Diagram

3.4.2 O-DU₈ Functional Module Description

Most of hardware modules in O-DU₈ are the same as the modules used by O-DU_{7.2}. Section 3.2.2 has the detailed description of those components. The differences between O-DU₈ and O-DU_{7.2} are the fronthaul interface and the physical layer functions that are performed within O-DU₈ (i.e., O-DU₈ performs all PHY functions vs O-DU_{7.2} only performs High PHY functions). In O-DU₈, CPRI interface is used for fronthaul. Figure 3-6 shows the various components used in O-DU₈, and additional functional modules are elaborated in the following:

CPRI Interface Device

This device provides the CPRI protocol used for the fronthaul interface.

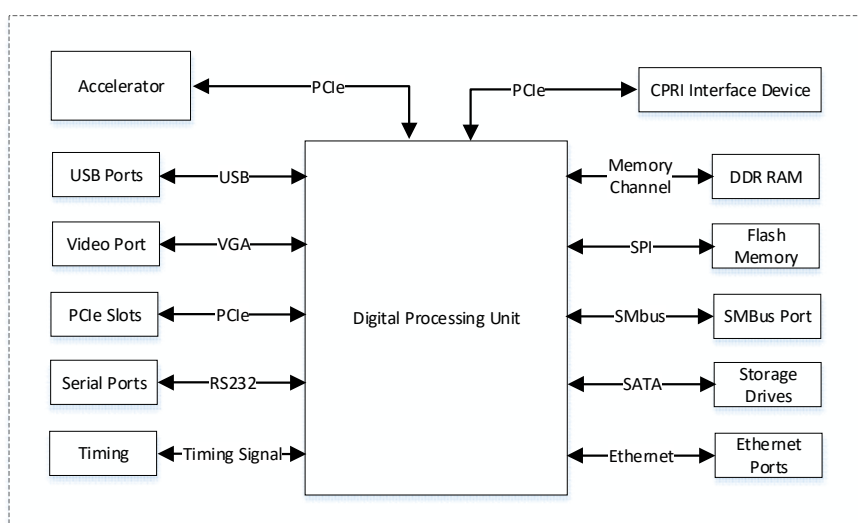


Figure 3-6: O-DU₈ Functional Block and Interface Diagram

3.4.3 O-DU₈ Interfaces

Among the O-DU₈ interfaces, CPRI is the fronthaul interface that is different from the O-DU₇₋₂ and is described below. The rest of the interfaces are the same as O-DU₇₋₂ interfaces which are described in section 3.2.3.

CPRI Interface

The CPRI interface shall comply with CPRI specification [5].

3.5 FHGW₇₋₂ - Option 7-2 to Option 7-2 Hardware Architecture

For the indoor picocell system, FHGW₇₋₂ performs the aggregation/distribution function for the uplink/downlink traffic to all the radio units. FHGW₇₋₂ has an upper interface which connects with O-DU₇₋₂, and a lower interface connected with O-RU₇₋₂. Here, both interfaces are based on the O-RAN WG4 defined fronthaul interface. FHGW₇₋₂ supports NR by default but LTE is not precluded.

3.5.1 FHGW₇₋₂ Architecture Diagram

Figure 3-7 depicts a FHGW₇₋₂ architecture diagram. The signal processing unit is the key component of the FHGW₇₋₂ which handles all the uplink and downlink traffic combining and distribution. The signal processing block can be either an FPGA or another digital processing unit. The back end and front end fronthaul interfaces are both Ethernet. In the front end direction, there is a broadcast function for connecting O-RU₇₋₂(s) and cascaded FHGW₇₋₂ with the same cell ID, and a demultiplex function for connecting O-RU₇₋₂(s) and FHGW₇₋₂ with different cell IDs. In the back end direction, all signals from connected O-RU₇₋₂(s) and FHGW₇₋₂ with same cell ID could be combined. See Figure below.

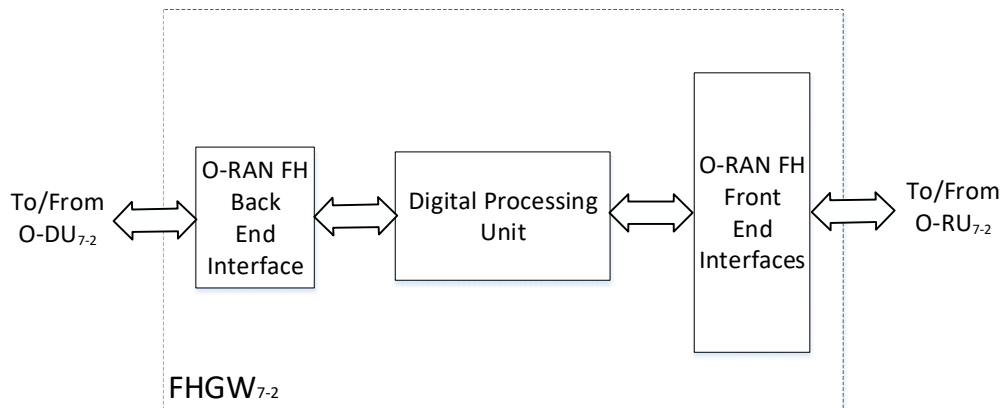


Figure 3-7: FHGW₇₋₂ Architecture Diagram

3.5.2 FHGW₇₋₂ Functional Module Description

The FHGW₇₋₂ functional module diagram is shown in Figure 3-8. The components include:

- Digital processing Unit: handles all the computation and signal processing functions.
- POE++: provides power over Ethernet.
- DC/DC: performs DC to DC conversion.
- CLK: provides the clock signals to the FHGW₇₋₂

- Memory: on board memory for data storage
- SPI: interface to connect with flash device for software and configuration data
- Debug Interface: interface is used for debugging purposes
- Ethernet: interfaces to connect with O-RU₇₋₂ And O-DU₇₋₂

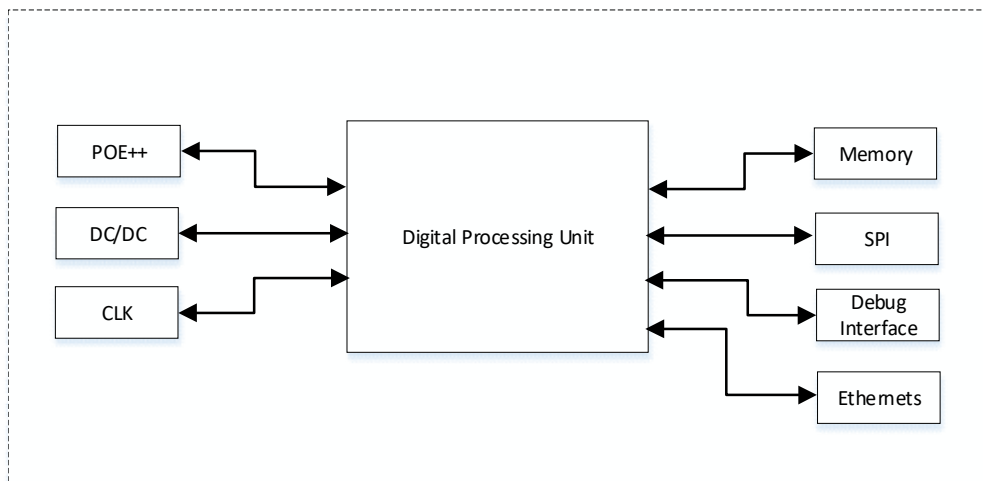


Figure 3-8: FHGW₇₋₂ Functional Module Diagram

3.6 FHGW₆ - Option 6 to Option 6 Hardware Architecture

No FHGW is needed for this indoor deployment scenario when adopting split option 6.

3.7 FHGW_{7-2->8} - Option 7-2 to Option 8 Hardware Architecture

FHGW_{7-2->8} performs both radio traffic multiplexing and fronthaul protocol translation tasks.

3.7.1 FHGW_{7-2->8} Architecture Diagram

Figure 3-9 illustrates the FHGW_{7-2->8} architecture. For FHGW_{7-2->8}, the digital processing unit can be the same as described in Section 3.5.1. With FHGW_{7-2->8}, the back end interface is Ethernet, while the front end interfaces are CPRI. The digital processing unit block shall conduct the protocol translation between the front end and back end interfaces.

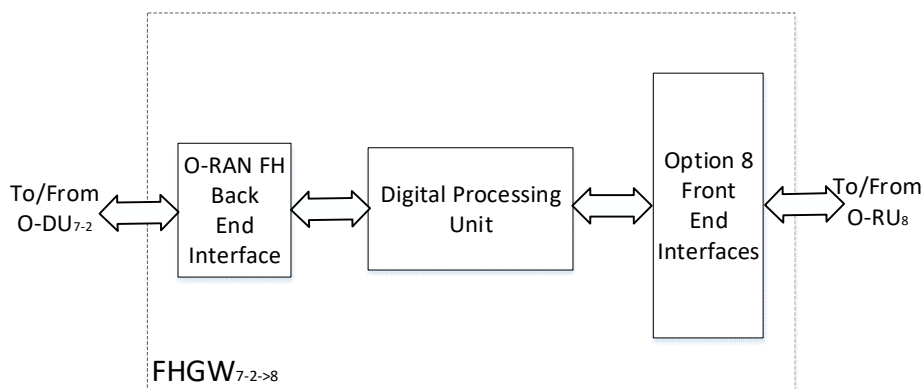


Figure 3-9: FHGW_{7-2->8} Architecture Diagram

3.7.2 FHGW_{7-2→8} Functional Module Description

Figure 3-10 is the FHGW_{7-2→8} functional module diagram. There is one 25G SFP+ port for back end link, one 25G SFP+ port for cascade link, and eight 10G CPRI ports for front end links. The front end ports also support remote power supply for O-RU₈(s) via POE++. In the forward direction, FHGW_{7-2→8} may connect an O-DU₇₋₂ to other O-RU₈(s) via a cascaded FHGW₈ supporting both the same and different cell IDs. However, it can also support demultiplexing functions for connecting O-RU₈(s) with different cell IDs. In the back end direction, all signals from connected O-RU₈(s) with the same cell IDs are combined via FHGW_{7-2→8} towards O-DU₇₋₂.

As different fronthaul interfaces are used by FHGW_{7-2→8}, it will translate the fronthaul protocol between front end and back end. FHGW_{7-2→8} shall also provide the low physical layer function.

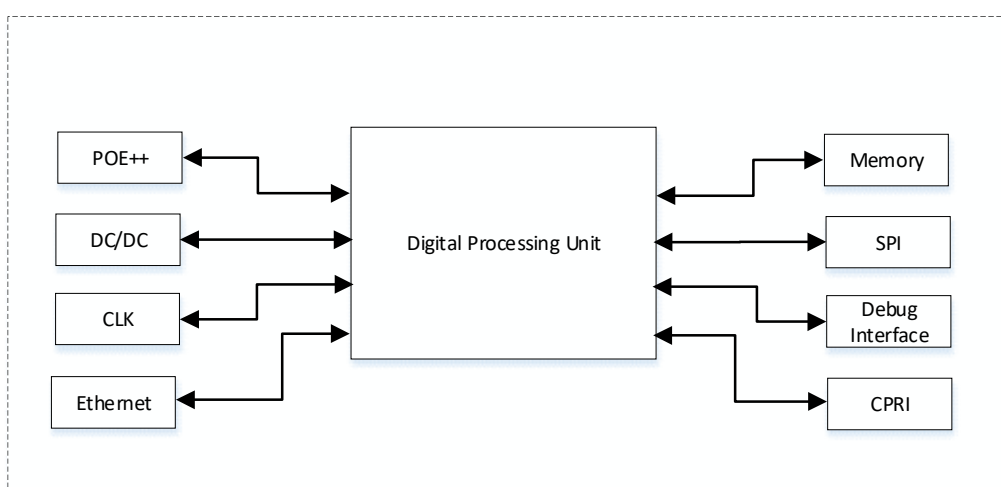


Figure 3-10: FHGW_{7-2→8} Functional Module Diagram

The FHGW_{7-2→8} functional module diagram is shown in Figure 3-10. The components include:

- Digital processing Unit: handles all the computation and signal processing functions.
- POE++: provides power over Ethernet.
- DC/DC: performs DC to DC conversion.
- CLK: provides the clock signals to the FHGW₇₋₂
- Memory: on board memory for data storage.
- SPI: interface to connect with flash device for software and configuration data.
- Debug Interface: The interface is used for debugging purposes.
- Ethernet: provides connection from O-DU₇₋₂ to FHGW_{7-2→8}
- CPRI: provides fronthaul connection from FHGW_{7-2→8} to O-RU₈

3.8 FHGW₈ - Option 8 to Option 8 Hardware Architecture

For option 8 to option 8 fronthaul gateway, both the front end and back end interfaces of FHGW₈ are based on CPRI.

3.8.1 FHGW₈ Architecture Diagram

Figure 3-11 illustrates the HW architecture FHGW₈. Note that the front end and the back end interfaces are both CPRI. The digital processing unit handles I/Q samples and all interface processing. FHGW₈ shall also

provide remote power supply functionality to the O-RU₈ and may be optionally cascaded with other(s) FHGW₈.

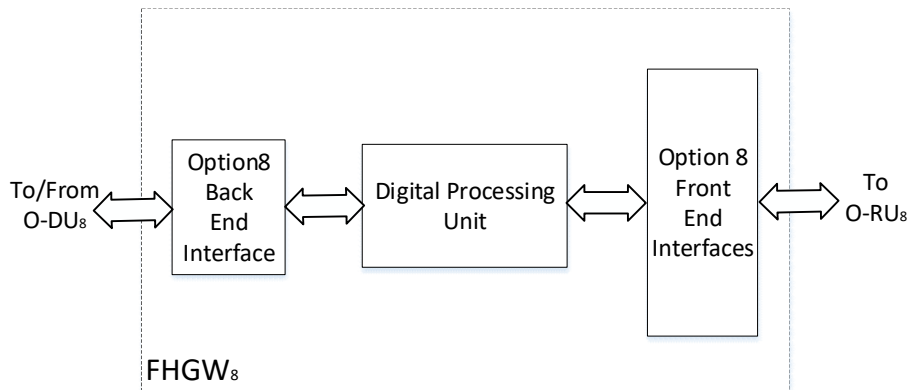


Figure 3-11: FHGW₈ Architecture Diagram

3.8.2 FHGW₈ Functional Module Description

Figure 3-12 illustrates FHGW₈ functional module diagram for an Option 8 CPRI switch. As shown in the FHGW₈ diagram, there is one 25G SFP+ port for back end link to O-DU₈, one 25G SFP+ port for cascade link, and eight 10G ports for front end links. All these ports support CPRI, while the front end ports further support remote power supply for O-RU₈ connection. In the front end direction, there is a broadcast function for connecting O-RU₈(s) and cascaded FHGW₈ with the same cell ID, and a demultiplex function for connecting O-RU₈(s) and FHGW₈ with different cell IDs. In the back end direction, all signals from connected O-RU₈(s) with the same cell ID could be combined to O-DU₈.

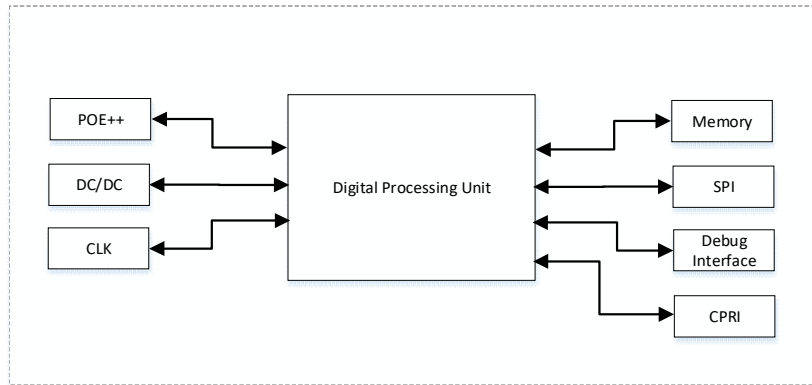


Figure 3-12: FHGW₈ Functional Module Diagram

The FHGW₈ functional module diagram is shown in Figure 3-12. The components include:

- Digital processing Unit: handles all the computation and signal processing functions.
- POE++: provides power over Ethernet
- DC/DC: performs DC to DC conversion.
- CLK: provides the clock signals to the FHGW₇₋₂
- Memory: on board memory for data storage.
- SPI: interface to connect with flash device for software and configuration data.
- Debug Interface: interface used for debug.

- CPRI: provides connection from O-DU₈ to FHGW₈ and from FHGW₈ to O-RU₈.

3.9 O-RU₇₋₂ Hardware Architecture

O-RU₇₋₂ consists of three major units, namely digital processing unit, RF processing unit and a timing unit as shown in Figure 3-13. The Ethernet interface complies with O-RAN WG4 open fronthaul interface. O-RU₇₋₂ can directly connect with O-DU₇₋₂ or connect through a FHGW₇₋₂. While O-RU₇₋₂ HW supports NR by default, LTE is not precluded.

3.9.1 O-RU₇₋₂ Architecture Diagram

O-RU₇₋₂ HW architecture consists of digital processing unit which handles all digital signal and interface processing and a RF processing unit handling all analog. There will be a transceiver right after digital processing block which converts between digital signals and analog signals, as well as frequency mixing. Then the PA/LNA amplifies the RF signal, and the antenna will be used to transmit and receive signal over the air. There is at least one Ethernet port available which is used as the O-RAN fronthaul interface.

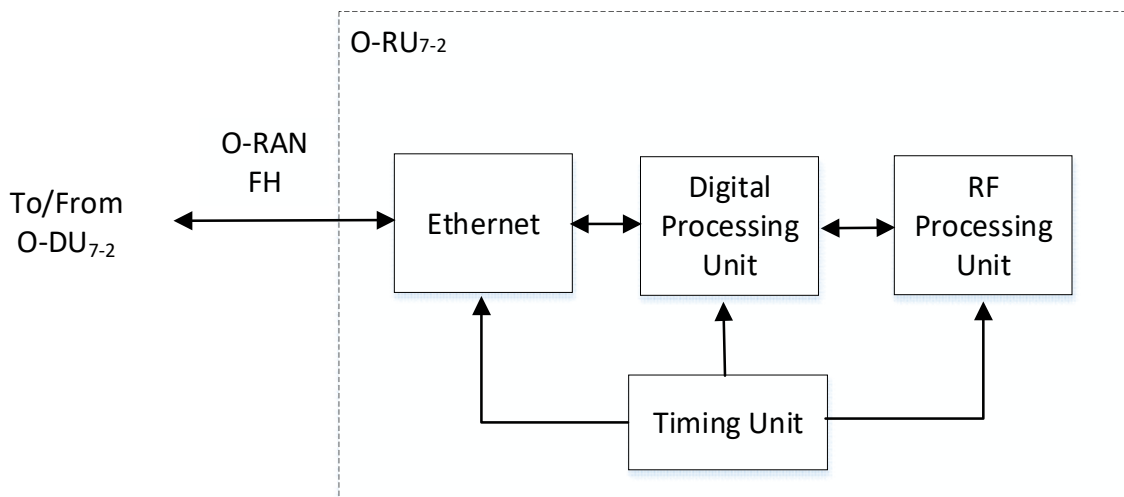


Figure 3-13: O-RU₇₋₂ Architecture Diagram

3.9.2 O-RU₇₋₂ Functional Module Description

Figure 3-14 illustrates the O-RU₇₋₂ functional module blocks that support O-RAN fronthaul with split option 7-2. There is at least one interface port which supports all fronthaul interface and PoE functionalities. The digital processing unit block of O-RU₇₋₂ is mainly responsible for low PHY functions such as FFT/iFFT, CP addition/removal, and PRACH filtering. Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD) are used for digital data processing. For bandwidth reduction, O-RU₇₋₂ architecture also supports the optional compression and decompression functions of FH interface. RF Processing Unit consists of the following blocks: Transceiver, Power Amplifier (PA)/ Low Noise Amplifier (LNA) and antenna. Transceiver is used for function of Analog to Digital Converter (ADC), Digital to Analog Converter (DAC) and mixer. The Timing unit may include Phase Locked Loop (PLL), local oscillator and other timing synchronization circuitries.

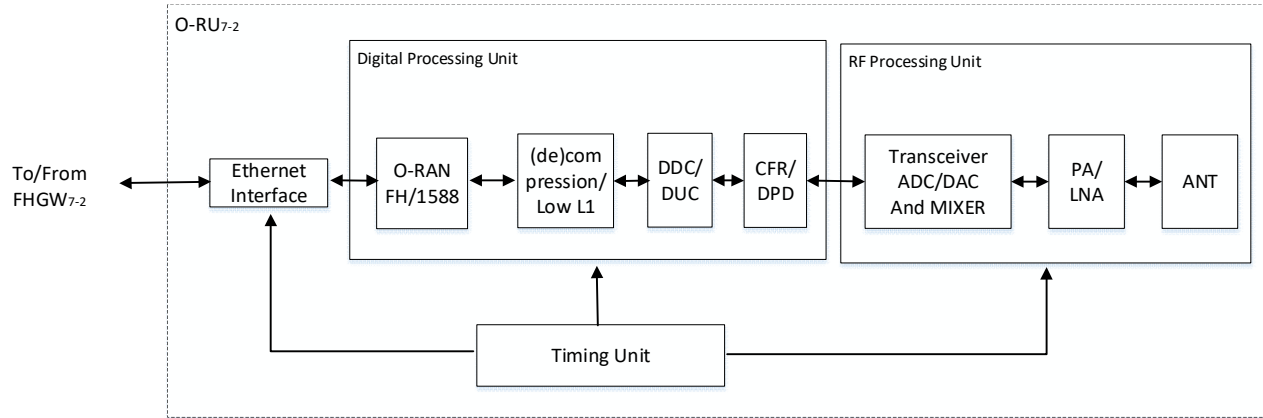


Figure 3-14: O-RU₇₋₂ Functional Module Diagram

3.10 O-RU₆ Hardware Architecture

O-RU₆ may be deployed in an integrated or split architecture, using the MAC/PHY split defined by the Small Cell Forum's FAPI or nFAPI interface [5].

3.10.1 O-RU₆ Architecture Diagram

O-RU₆ consists of three major Units, namely digital processing unit, RF processing unit and a timing unit as shown in Figure 3-15. Option 6 is referred to as interface between PHY and MAC as defined by the Small Cell Forum [6] and is carried by the fronthaul interface over Ethernet as shown in Figure 3-15. In the next section there is a brief functional description of modules within these major blocks of O-RU₆.

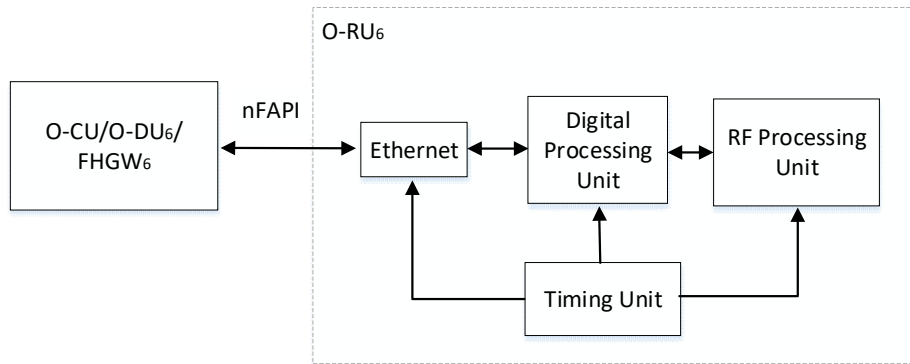


Figure 3-15: O-RU₆ Architecture Diagram

3.10.2 O-RU₆ Functional Module Description

Figure 3-16 shows the major components used in O-RU₆.

The digital processing unit handles transport/interface processing, complete PHY (low and high PHY functions) modem processing, Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD).

The RF processing unit is comprised of the transceiver block which is responsible for functions such as Analog to Digital Converter (ADC), Digital to Analog Converter (DAC) and mixer, an amplification block

which contains Power Amplifier (PA)/ Low Noise Amplifier (LNA), etc and antenna subsystem that captures and/or transmits radio electromagnetic waves. O-RU₆ may use Power over Ethernet (PoE) to simplify deployment complexity.

The Timing Unit may include Phase Locked Loop (PLL), local oscillator and timing synchronization circuitry.

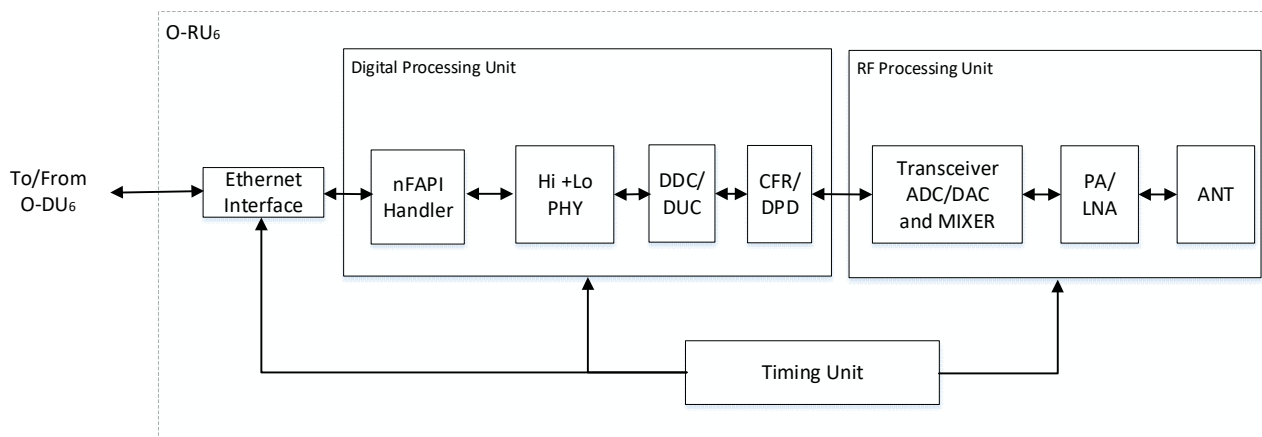


Figure 3-16: O-RU₆ Functional Module Diagram

3.11 O-RU₈ Hardware Architecture

O-RU₈ hardware architecture is similar to O-RU₇₋₂ as described in the previous section. The major functional differences reside in the fronthaul interface functions and Physical layer functions. Note that in split option 8 architecture, all PHY functions are performed in the O-DU₈. The digital signal processing block will comprehend the functional differences and make adjustment accordingly. The RF processing blocks are the same in this case.

3.11.1 O-RU₈ Architecture Diagram

Figure 3-17 shows a complete O-RU₈ architecture diagram. CPRI is adopted as the fronthaul interface in O-RU₈. The CPRI interface shall be included as part of the white box reference design document. The programmable digital signal processing block handles all I/Q data samples processing, management and control functions. The transceiver block does the radio signal conversion from analog to digital and vice versa.

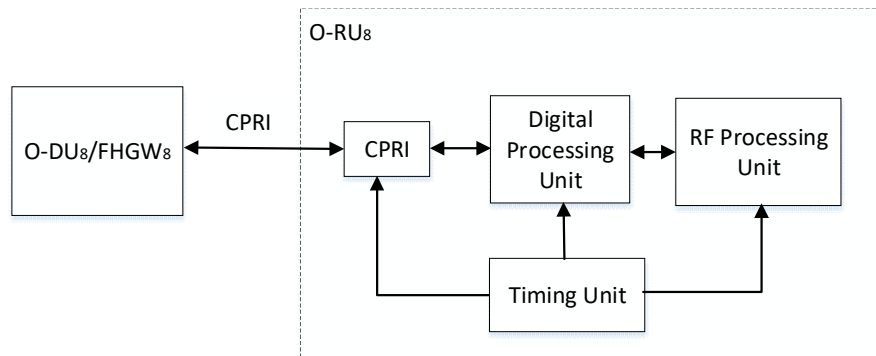


Figure 3-17: O-RU₈ Architecture Diagram.

3.11.2 O-RU₈ Functional Module Description

Figure 3-18 depicts functional block diagram of one example of O-RU₈ which supports split architecture with Option 8. The CPRI interface shall be included as part of the Whitebox reference design document. The programmable digital signal processing block handles all I/Q data samples processing, management and control functions. Therefore, within O-RU₈, there shall be at least copper or fiber interface to support transmission of CPRI protocol. The digital processing unit of O-RU₈ is responsible for CPRI protocol processing, Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD). The RF processing unit is the same as O-RU₇₋₂, and O-RU₆. The Transceiver block comprises of Analog to Digital Converter (ADC), Digital to Analog Converter (DAC) and mixer. The analog blocks include: Power Amplifier (PA)/ Low Noise Amplifier (LNA), etc. The Timing Unit may include Phase Locked Loop (PLL), local oscillator and other timing synchronization circuitries.

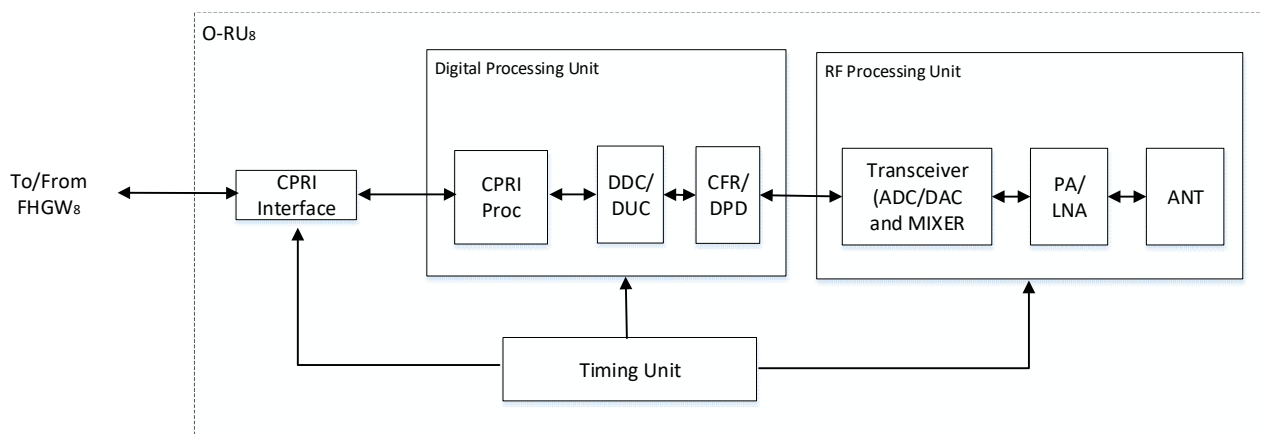


Figure 3-18: O-RU₈ Functional Module Diagram

3.12 Integrated gNB-DU Hardware Architecture

Integrated gNB-DU is a platform that where all functionalities of an O-DU and an O-RU are performed in the same Whitebox. The gNB-DU's hardware will support all Layer 1 and Layer 2 functionalities of the base station.

3.12.1 Integrated gNB-DU Architecture Diagram

Figure 3-19 shows the gNB-DU architecture block diagram. The integrated gNB-DU consists of a digital processing unit, a RF processing unit, and a timing unit. The gNB-DU connects to O-CU through an F1 interface. The contents of the processing units are described next.

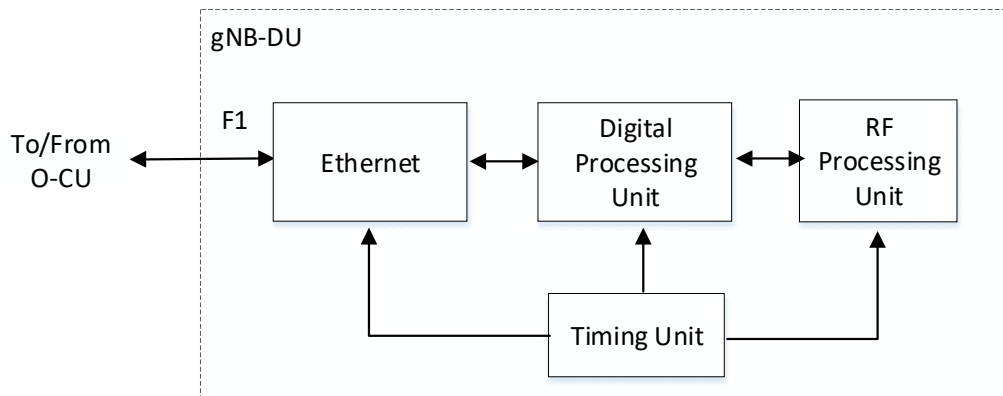


Figure 3-19: Integrated gNB-DU Architecture Diagram

3.12.2 Integrated gNB-DU Function Module Description

Figure 3-20 illustrates the gNB-DU function modules diagram. The Ethernet port is used as the interface to the O-CU via an F1 interface. The digital processing unit consists of RLC/MAC processing, physical layer processing, ADC/DAC blocks and CFR/DPD blocks. The RF processing unit includes the transceiver block, ADC/DAC, mixers, PA/LNA/RF filters and Antennas. The timing unit includes PLL, local oscillators, and other timing and synchronization circuitry. Hardware interfaces that are used internally may include PCIe, SPI, JESD, etc.

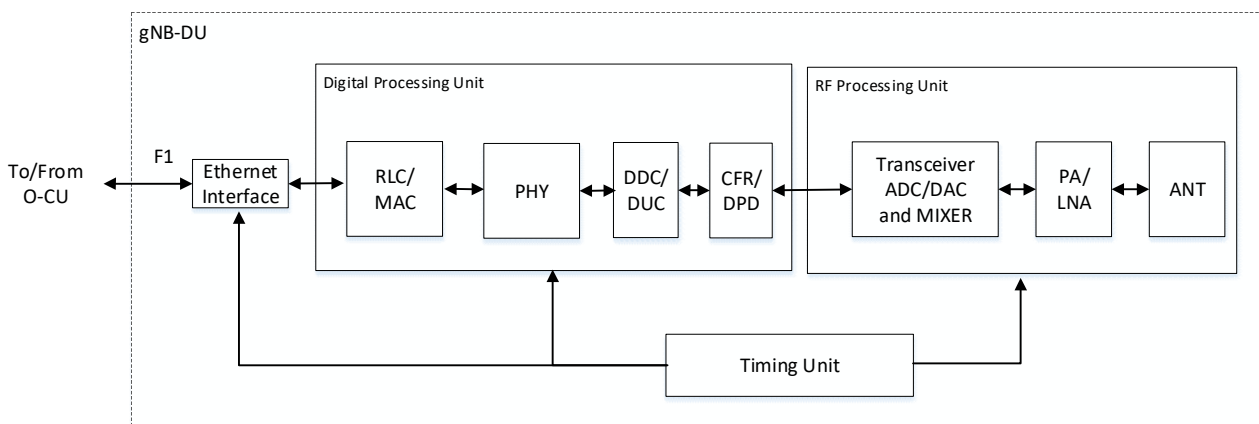


Figure 3-20: Integrated gNB-DU Functional Module Diagram

3.13 AIO White Box Hardware Architecture

Based on the gNB physical implementation architectures discussed earlier, this section provides the architecture, major building blocks and all external interfaces for the AIO Whitebox.

3.13.1 AIO Architecture Diagram

The AIO whitebox hosts L1/L2/L3 functions and radio frequency functions, and all the functions are implemented on the same hardware platforms. Figure 3-21 shows the hardware blocks and external interfaces of AIO Whitebox.

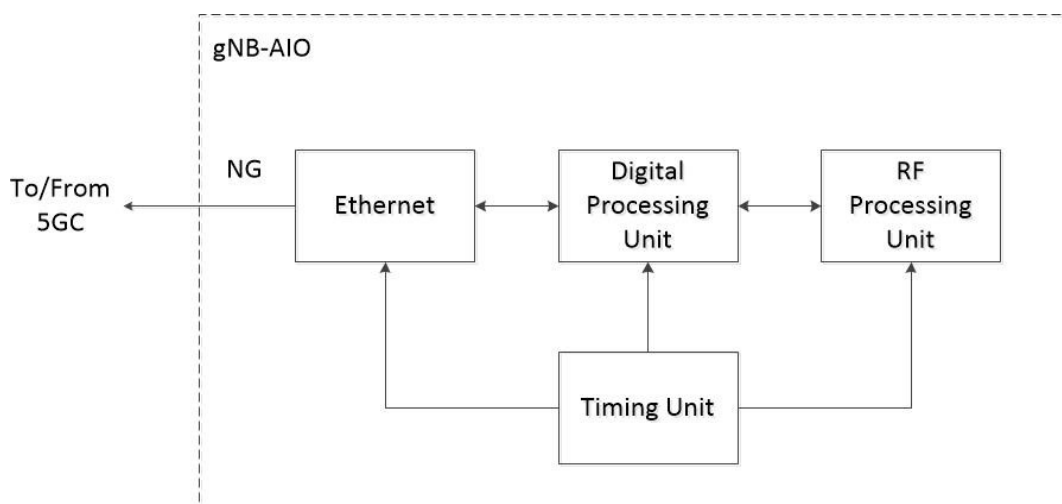


Figure 3-21: AIO White box Architecture Diagram

3.13.2 AIO Functional Module Description

The AIO whitebox functional architecture comprises Digital Signal Processing, Connectivity (GbE) units, RF processing unit and Timing unit, as well as a timing unit as shown in Figure 3.13. The descriptions and requirements of functional module are as follows:

Digital Processing Unit

The digital Processing unit can be any GPP, FPGA or digital signal processor (DSP), with I/O chipset. It may also be in the form of System-On-Chip (SOC), or Multi-Chip Package (MCP).

The digital Processing unit hosts L1/L2/L3 processing, ADC/DAC blocks and CFR/DPD blocks. The ADC and DAC are mixed signal devices responsible for conversion of data between the digital and analog domains. As such, this block can be included as part of the either the digital processing unit or the RF processing unit.

RF Processing Unit

The RF Processing Unit consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low Noise Amplifier (LNA) and TX/RX filters.

1 **Timing Unit**

2 The timing unit includes any clock and frequency synthesis required as well as other timing and
3 synchronization circuits.

4 **Ethernet Controller**

5 The Ethernet ports transport the backhaul traffic according to the gNB hardware node requirements. The
6 Ethernet device shall support GNSS/BDS based timing synchronization.

7 **3.13.3 AIO Interfaces**

8 The AIO whitebox interfaces through backhaul with the 5GC core network. The backhaul interfaces are
9 typically implemented with GbE transport/connections.

Chapter 4 White Box Hardware Requirements

This chapter provides the requirements for various white boxes used within the indoor picocell base station. These white boxes are O-CU, O-DU_x, O-RU_x and FHGW_x. The O-CU and O-DU_x can be implemented in an integrated fashion into one white box hardware or they can be separated. The reference design based on specification shall meet all these requirements.

4.1 O-CU Requirements

O-CU requirements are described in the following sections, which include the performance, interface, environmental, EMC, mechanical, thermal and power requirements.

4.1.1 O-CU Performance

The performance requirements of the O-CU are listed in Table 4-1.

Table 4-1 :O-CU Performance Requirements

Parameter	Requirement	Description	Priority
Synchronization	Support GPS Synchronization; Support 1588V2 Synchronization Support BeiDou Synchronization Support BeiDou and GPS switching	Timing synchronization method	High
Supported DU Number	For O-CU/O-DU _x integrated architecture: 1	Number of O-DU _x (s) connected to a single O-CU	High
	For O-CU/O-DU _x non-integrated architecture: at least 8		High
Supported Cell Number	For O-CU/O-DU _x integrated architecture: at least 4	Cells supported by an O-CU	High
	For O-CU/O-DU _x non-integrated architecture: at least 32		High
Supported RRC link number	For O-CU/O-DU _x integrated architecture: at least 4800	RLC links supported by O-CU	High
	For O-CU/O-DU _x non-integrated architecture: at least 1280		High
Throughput between O-DU _x	For O-CU/O-DU _x integrated architecture: at least 6.5Gbps	Throughput between O-CU and O-DU _x link	High
	For O-CU/O-DU _x non-integrated architecture: at least 16Gbps		High
Latency	TBD	O-CU latency	High

4.1.2 O-CU Interfaces

The interface requirements of the O-CU are listed in Table 4-2: O-CU Interface Requirements.

Table 4-2: O-CU Interface Requirements

Parameter	Requirement	Description	Priority
Transport Interface	At least 25 GbE F1 Interface to connect with O-DU _x ^(a)	O-CU transport links to O-DU _x and 5GC	High
	At least 10 GbE NG interface to connect with 5G core ^(b)		High
	At least 25 GbE NG interface to connect with 5G core ^(a)		High

Note:

(a) This requirement is for separated architecture.

(b) This requirement is for integrated architecture. For O-CU and O-DU_x integrated solution, the F1 interface is an internal logic interface.

4.1.3 O-CU Environmental and EMC

The EMC requirements of the O-CU are listed in Table 4-3.

Table 4-3: O-CU EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Complying with the requirements of 3GPP TS 38.113 (2017-12R15) [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

4.1.4 O-CU Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-CU are listed in the following tables.

Table 4-4: O-CU Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimension	Built in any 19" standard rack, or stand alone, with overall height of no more than 5U ^(a) , the depth (including the connector) must be less than 750mm.	Measurement in three dimensions	High

Status Indicator LED	At least include the following status indicators: 1 indicating on/off status of the power supply 1 indicating on/off status of the transmission link	Indicator light	High
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Note:

- (a) Note that this dimension is application dependant and its value may change accordingly.

Table 4-5: O-CU Thermal Requirements

Parameter	Requirement	Description	Priority
Reliability	Work steadily and reliably over a long period of time under the following environmental conditions: Operating Temperature: - 5 °C ~ + 55 °C Relative humidity: 15% ~ 85%	Environmental requirements for reliability	High

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

Table 4-6: O-CU Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300v, frequency range of 45Hz ~ 65Hz.	Power supply for O-CU	High
Power dissipation	< 300W full load operation	O-CU Power Requirement	High

Note: For O-CU and O-DU_x integrated solution, this requirement for O-CU portion is captured in the following O-DU_x section.

4.2 O-DU_x Common Requirements

The O-DU_x here takes the function of high PHY or whole PHY, OAM function and layer 2. Usually the hardware is placed in the machine room, which can be collocated in the coverage building or in the central machine room far away from the coverage building.

4.2.1 O-DU_x Performance

The performance requirements of the O-DU_x are listed in Table 4-7.

Table 4-7: O-DU_x Performance Requirements

Parameter	Requirement	Description	Priority
Channel Bandwidth	Up to 100MHz (DL+UL)	RF Bandwidth	High
Antenna Number	2T2R	Tx/Rx antenna number	High
	4T4R		High
Transmission distance	Directly connected with Fronthaul Gateway ≥10Km	Distance between DU and FHGW _x /RU	High
Connect Topology	Support at least two Fronthaul Gateways of chain topology connections	Fronthaul Gateway connection topology	High
Synchronization	Support GPS Synchronization; Sync. between BS <±1.5us; Carrier freq. error within one subframe <±0.1 PPM Support 1588V2 Synchronization Support BeiDou Synchronization Support BeiDou and GPS switching	Timing synchronization method	High
Capacity MIMO	Support at least 4 100MHz bandwidth 2T2R cells	MIMO related capability	High
	Support at least 2 100MHz bandwidth 4T4R cells		Low
	Support at least 8 100MHz bandwidth 2T2R cells		Medium
	Support at least 4 100MHz bandwidth 4T4R cells		High
	DL:2*2MIMO UL:2*2MIMO		High
	DL:4*4MIMO UL:2*2MIMO		High
Peak Rate	With 100MHz bandwidth and 74% for DL, the DL peak throughput of one cell shall not be lower than 850Mbps for 2*2 or 1700Mbps for 4*4, and the UL peak throughput of one cell shall not be lower than 190Mbps.2T2R)	Peak data rate	High
Modulation	DL: QPSK,16QAM,64QAM,256QAM UL: $\pi/2$ -bpsk, QPSK, 16QAM, 64QAM, 256QAM	Modulation schemes	High

Latency	Control Plane<20ms, (def: message 1 to message 5) User Plane DL <4ms, UL<6ms (def: PDCP SDU-> PDCP SDU)	Control/user plane Latency	High
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4.2.2 O-DU_x Interfaces

The interface requirements of the O-DU_x are listed in Table 4-8.

Table 4-8: O-DU_x Interface Requirements

Parameter	Requirement	Description	Priority
Transport Interface	Fronthaul interfaces connected with either radio unit or Fronthaul Gateway	O-DU _x transport links	High
	At least one NG interface to 5GC		High
	At least 10 GbE F1 interface to connect with O-CU ^(a)		High

Notes:

(a) For O-CU and O-DU_x that are separated physically

4.2.3 O-DU_x Environmental and EMC

The EMC requirements of the O-DU_x are listed in Table 4-9.

Table 4-9: O-DU_x EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Complying with the requirements of 3GPP TS 38.113 (2017-12R15) [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

4.2.4 O-DU_x Mechanical, Thermal and Power

The mechanical requirements of the O-DU_x are listed in Table 4-10.

Table 4-10: O-DU_x Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimension	Built in any 19" standard rack, or stand alone, and the height does not exceed 2U ^(a) , the depth (including the connector) must be less than 450mm.	Measurement in three dimensions	High

Status Indicator LED	At least includes the following status indicators: 1 indicating the on/off status of optical fiber interface 1 indicating off status of the power supply 1 indicating on/off status of the transmission link	Indicator light	High
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Note:

(a) Note that this dimension is application dependant and its value may change accordingly.

The thermal requirements of the O-DU_x are listed in Table 4-11.

Table 4-11: O-DU_x Thermal Requirements

Parameter	Requirement	Description	Priority
Reliability	Work steadily and reliably over a long period of time under the following environmental conditions: Operating Temperature: - 5 °C ~ + 55 °C Relative humidity: 15% ~ 85%	Environmental requirements for reliability	High

The power requirements of the O-DU_x are listed in Table 4-12.

Table 4-12: O-DU_x Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	DC -48 VDC (-40v ~ -57v) (can be connected to AC/DC converter) or AC 220V power supply, voltage range of 140V~ 300v, frequency range of 45Hz ~ 65Hz.	Power supply for O-DU _x	High
Power Dissipation	< 500W, with 4 cells full load operation	O-DU _x Power Requirement	High
	< 350W, with 4 cells full load operation		High
	< 800W, with 8 cells full load operation		Medium

4.3 O-DU_x Split Option Specific Requirements

In addition to the common O-DU_x requirements, there may be some specific requirements that apply to O-DU_x due to the split option. These requirements are listed in the following sections.

4.3.1 O-DU₇₋₂ Specific Requirements

The specific requirements of O-DU₇₋₂ are listed in Table 4-13.

Table 4-13: O-DU₇₋₂ Specific Requirements

Parameter	Requirement	Description	Priority
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Transport Interface	At least 4 Ethernet interfaces connected with Fronthaul Gateway	O-DU ₇₋₂ fronthaul transport links	High
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4.3.2 O-DU₆ Specific Requirements

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The specific requirements of O-DU₆ are listed in Table 4-14.

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Table 4-14: O-DU₆ Specific Requirements

Parameter	Requirement	Description	Priority
Transport Interface	At least one Ethernet interfaces for fronthaul	O-DU ₆ fronthaul transport links	High

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4.3.3 O-DU₈ Specific Requirements

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The specific requirements of the O-DU₈ are listed in Table 4-15.

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Table 4-15: O-DU₈ Specific Requirements

Parameter	Requirement	Description	Priority
Transport Interface	At least 4 CPRI (option 8) interfaces connected with Fronthaul Gateway	O-DU ₈ fronthaul transport links	High

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4.4 O-RU_x Common Requirements

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The O-RU_x common requirements apply to all the radio units regardless of the split option. In the indoor environment, O-RU_x hardware is placed on the cell or wall of the coverage building; it converts the base band signal into RF signal or vice versa to supply the coverage.

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4.4.1 O-RU_x Performance

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The O-RU_x performance requirements cover all the aspects of radio unit including frequency bands, antenna configurations, power efficiency, etc. Table 4-16 lists the performance parameters related to O-RU_x.

Table 4-16: O-RU_x Performance Requirements

Parameter	Requirement	Description	Priority
Operating band	n2, n4, n5, n13, n41, n48, n66, n77, n78, n79	Radio frequency band	High
Channel bandwidth	Up to 100MHz (DL+UL)	Frequency bandwidth	High
Transmitter and receiver number	2T2R	Tx/Rx Antenna numbers	High
	4T4R		High
Output Power Accuracy	Under normal condition: $\pm 2\text{dB}$	Power accuracy	High
Tx off Power Level	less than -89dBm/MHz	Tx power level	High
EVM at maximum output power	64QAM: EVM smaller than 5% 256QAM: EVM smaller than 3.5%	Max output power	High
Operating band unwanted emissions	The Operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.4.2.4 in 3GPP TS 38.104 [3].	RF operation band unwanted emissions requirement	High
Transmitter spurious emissions	The Operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.4.2.4 in [3].	Deliberately generated RF signal by transmitter	High
receiver sensitivity	The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel of G-FR1-A1-5, the reference sensitivity levels should not higher than -94dBm.	The weakest signal receiver is able to identify and process	High
Blocking	In Channel selection, ACS, In-band blocking, out-band blocking, IMD and other receiver specification must follow the 3GPP guidelines in [3], the reference sensitivity is allowed to degrade at most 6dB under all kinds of interference signal and corresponding level.	Channel selection related requirement	High
Other specifications	Except for all the RF specifications listed above, other RF specifications must follow the requirement in [3].	Addition standard to comply	High
Downlink modulation Mode	QPSK、16-QAM、64-QAM、256-QAM	DL Modulation schemes	High
Uplink modulation mode	$\pi/2$ -BPSK、QPSK、16-QAM、64-QAM、256-QAM	UL Modulation schemes	High
	256QAM		medium

Output power	The rated output power of the O-RU _x is 0.5 W. 2T2R	RF radiation power	High
	The rated output power of the O-RU _x is 1 W. 2T2R		High
	The rated output power of the O-RU _x is 1 W. 4T4R		High

4.4.2 O-RU_x Interfaces

The interface related requirements of the O-RU_x are listed in Table 4-17.

Table 4-17: O-RU_x Interface Requirements

Parameter	Requirement	Description	Priority
Interfaces Number	The O-RU _x must have at least one fronthaul interface based on the split option supported	Number of fronthaul links	High

4.4.3 O-RU_x Environmental and EMC

The environmental and EMC requirements of the O-RU_x are listed in Table 4-18.

Table 4-18: O-RU_x Environmental and EMC Requirements

Parameter	Requirement	Description	Priority
Mounting method	Wall and ceiling mounting.	Mounting requirement	High
Grounding	The O-RU _x must support Joint Grounding Method and can working normally when the grounding resistor is less than 10Ω.	Grounding requirement	High
EMC	Complying with the requirements of 3GPP TS 38.113 [7]	Electromagnetic Compatibility requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

4.4.4 O-RU_x Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the O-RU_x are listed in Table 4-19.

Table 4-19: O-RU_x Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Weight	The gross weight of the O-RU _x must smaller than 3kg.	Weight requirement	High
Dimension	The dimension of the O-RU _x must smaller than 3L.	Measurement in dimension	High
Stability	The failure rate of O-RU _x must not exceed 2%	Stability requirement	High
Power Consumption	At full load, the power consumption must not exceed 40W. 2T2R	Power requirement	High
	At full load, the power consumption must not exceed 50W. 4T4R		High
Power supply	O-RU _x must support either an isolated POE or isolated optical fiber with integrated power cable as the power supply source	Power Support Requirement	High
Level of protection	The protection level of O-RU _x is equivalent to the IP31 standard.	Protection level	High
Temperature and moisture	The O-RU _x must be operated and stored under those conditions: Temperature : -5°C~+55°C moisture : 5%~95%	Ambient temperature and moisture requirement	High
Atmospheric pressure	The O-RU _x must operate normally under the atmospheric pressure between 70 to 106Kpa.	Operation atmospheric pressure requirement	High
Cooling mode	Natural heat dissipation.	System cooling requirement	High

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4.5 O-RU_x Split Option Specific Requirements

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Besides the common requirements which shall apply to all the radio unit types. The following sections list all the specific requirements that only apply to the designated split option.

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4.5.1 O-RU₇₋₂ Specific Requirements

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The O-RU₇₋₂ must have one RJ45 or SFP 10G Ethernet interface used as fronthaul interface. The O-RU₇₋₂ must support the lower physical layer functions and comply with [4].

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4.5.2 O-RU₆ Specific Requirements

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The O-RU₆ must have at least one RJ45 10 gigabit per second (Gbps) Ethernet interface or at least one 10 Gbps optical interface. In some cases, depending on the fronthaul throughput associated with smaller air

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interface bandwidths, a 1 or 2.5 Gbps Ethernet or optical interface may be sufficient, depending on system operator requirements. SCF 5G nFAPI should be supported.

At full load, the power consumption must not exceed the limit allowed by IEEE802.3at/PoE+, which is 25.5W.

4.5.3 O-RU₈ Specific Requirements

The O-RU₈ must have at least one SFP 10Gbps optical interface for CPRI interface. The O-RU₈ fronthaul interface shall comply with [5].

4.6 FHGW_x – Common Requirements

A Fronthaul gateway is used in many scenarios as the aggregation point of radio units, as well as to provide power to the O-RUs. The FHGW_x usually is deployed close to O-RUs (e.g. indoor coverage: deployed on the same floor or in the same building with multiple radios less than 100 meters). FHGW_x distributes traffic from the O-DU_x to each O-RU_x, and it combines the uplink traffic from all radios before sending it to the O-DU_x. The benefit of using the FHGW_x includes radio traffic aggregation and centralized power supplied to radio units. When different fronthaul interfaces are used by the O-DU_x and O-RU_x, the FHGW_x can act as a fronthaul protocol converter, e.g. from eCPRI to CPRI.

4.6.1 FHGW_x Performance

The performance requirements of the FHGW_x are listed in Table 4-20.

Table 4-20: FHGW_x Performance Requirements

Parameter	Requirement	Description	Priority
Transmission distance	The transmission distance must not less than 100m when using POE. The transmission distance must not less than 200m when using optical fiber with power cable	Wired line distance limit	High

4.6.2 FHGW_x Interfaces

The interface requirements of the FHGW_x are listed in Table 4-21.

Table 4-21: FHGW_x Interface Requirements

Parameter	Requirement	Description	Priority
Interface	FHGW _x must have at least 2 fronthaul interfaces, one for the connection with O-DU _x , the other one for the connection with the second FHGW _x .	Number of Fronthaul links	High

	FHGW _x must support fronthaul connections for 8 O-RU _x and must supply the power to the O-RU _x through POE or optical fiber with power cable.		High
Cascade	Each FHGW _x must support a cascade connection with another fronthaul gateway.	Topology between FHGW _x (s)	High

4.6.3 FHGW_x Environmental, EMCs

The EMC requirements of the FHGW_x are listed in Table 4-22.

Table 4-22: FHGW_x EMC Requirements

Parameter	Requirement	Description	Priority
Mounting method	Wall and ceiling mountable.	Mounting method	High
Grounding	The FHGW _x must support Joint Grounding Method and can work normally when the grounding resistor is less than 10Ω.	FHGW _x grounding requirement	High
EMC	Complying with the requirements in [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

4.6.4 FHGW_x Mechanical, Thermal and Power

The mechanical, thermal and power requirement of the FHGW_x are listed in Table 4-23.

Table 4-23: FHGW_x Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Power Consumption	The static power consumption (not supplying power to another FHGW _x) of each fronthaul gateway must not exceed 55W.	Power requirement	High
Dimension	Each FHGW _x can be mounted on the 19-inch rack. The height must not exceed 1U.	Measurement in dimensions	High
Noise	Under Normal temperature (25°C) condition, the noise level must smaller than 40dBA. Under extreme condition (40°C), the noise level must smaller than 45dBA.	Noise level requirement	High

Temperature and moisture	The FHGW _x must be operated and stored under these conditions: Temperature : -5°C ~ +55°C moisture : 15%~85%	Ambient temperature and moisture	High
Atmospheric pressure	The FHGW _x must operate normally under the atmospheric pressure between 70 to 106Kpa.	Operational atmospheric pressure requirement	High

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4.7 FHGW_x – Split Option Specific Requirements

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For architecture with different split options, FHGW_x may have specific hardware requirements that only apply to a designated type of fronthaul protocol. Those requirements are listed in the following sections.

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Note that Split option 6 does not require any FHGW.

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4.7.1 FHGW₇₋₂ Specific Requirements

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FHGW₇₋₂ must have at least two 10 Gbps optical or RJ45 Ethernet interfaces, one for the connection with O-DU₇₋₂, the other for the connection with another FHGW₇₋₂.

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Each FHGW₇₋₂ must support the connection to 8 O-RU₇₋₂ and supply the power to the O-RU₇₋₂ through POE or optical fiber with power cable.

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4.7.2 FHGW_{7-2->8} Specific Requirements

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FHGW_{7-2->8} must have at least two 10 Gbps optical or RJ45 Ethernet interfaces, one for the connection with O-DU₇₋₂, the other for the connection with another FHGW_{7-2->8}.

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Each FHGW_{7-2->8} must support 8 connection links O-RU₈ and supply the power to the O-RU₈ through optical fiber with power cable.

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4.7.3 FHGW₈ Specific Requirements

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FHGW₈ must have at least 2 optical interfaces, one for the connection between O-DU₈, the other for the connection with another FHGW₈.

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Each FHGW₈ must support the 8 CPRI links for O-RU₈ and supply the power to O-RU₈ via optical fiber with power cable.

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4.8 Integrated gNB-DU Requirements

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The following sections list the requirements for the integrated gNB-DU which includes performance, interfaces, mechanical, thermal, etc.

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4.8.1 Integrated gNB-DU Performance

Table 4-24 lists the performance requirements for gNB-DU.

Table 4-24: Integrated gNB-DU Performance Requirements

Parameter	Requirement	Description	Priority
Operating band	n2, n4, n5, n13, n41, n48, n66, n77, n78, n79,	Radio frequency band	High
Channel bandwidth	Up to 100MHz (DL+UL)	Frequency bandwidth	High
Transmitter and receiver number	2T2R	Tx/Rx Antenna numbers	High
	4T4R		High
Output Power Accuracy	Under normal condition: $\pm 2\text{dB}$	Power accuracy	High
Tx off Power Level	Less than -89dBm/MHz	Tx power level	High
EVM at maximum output power	64QAM: EVM smaller than 5% 256QAM: EVM smaller than 3.5%	Max output power	High
Operating band unwanted emissions	The Operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.4.2.4 in [3].	RF operation band unwanted emissions requirement	High
Transmitter spurious emissions	The Operating band unwanted emissions must satisfy the Category B limit defined by the section 6.6.4.2.4 in [3].	Deliberately generated RF signal by transmitter	High
receiver sensitivity	The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel of G-FR1-A1-5, the reference sensitivity levels should not higher than -94dBm .	The weakest signal receiver is able to identify and process	High
Blocking	In Channel selection, adjacent channel selectivity (ACS), In-band blocking, out-band blocking, Intermodulation Distortion (IMD) and other receiver specification must follow [3]; the reference sensitivity is allowed to degrade at most 6dB under all kinds of interference signal and corresponding level.	Channel selection related requirement	High

Other specifications	Except for all the RF specifications listed above, other RF specifications must follow the requirement in [3].	Addition standard to comply	High
Downlink modulation Mode	QPSK、16-QAM、64-QAM、256-QAM	DL Modulation schemes	High
Uplink modulation mode	$\pi/2$ -BPSK、QPSK、16-QAM、64-QAM	UL Modulation schemes	High
	256QAM		Medium
Output power	The rated output power of the O-RU _x is 0.5W. 2T2R	RF radiation power	High
	The rated output power of the O-RU _x is 1W. 2T2R		High
	The rated output power of the O-RU _x is 1W. 4T4R		Low
Synchronization	Sync. between BS $\leq \pm 1.5\mu s$; Carrier freq. error within one subframe $\leq \pm 0.1$ PPM Support 1588V2 Synchronization Support BeiDou Synchronization Support GPS Synchronization; Support BeiDou and GPS switching	Timing synchronization method	High

4.8.2 Integrated gNB-DU Interfaces

Table 4-25 lists the Interface Requirement for integrated gNB-DU.

Table 4-25: Integrated gNB-DU Interface Requirements

Parameter	Requirement	Description	Priority
Interfaces Number	The gNB-DU must have at least one Ethernet interface for F1 interface	Number of backhaul links	High

4.8.3 Integrated gNB-DU Environmental, EMC

Table 4-26 lists the Environmental and electromagnetic compatibility (EMC) related requirements for integrated gNB-DU.

Table 4-26: Integrated gNB-DU Environmental and EMC Requirements

Parameter	Requirement	Description	Priority
Mounting method	Wall and ceiling mounting.	Mounting requirement	High
Grounding	The gNB-DU must support Joint Grounding Method and can operate normally when the grounding resistor is less than 10Ω.	Grounding requirement	High
EMC	Complying with the requirements of 3GPP TS 38.113 [7].	Electromagnetic Compatibility requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

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4.8.4 Integrated gNB-DU Mechanical, Thermal and Power

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Table 4-27 lists the mechanical, thermal and power requirements for integrated gNB-DU.

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Table 4-27: Integrated gNB-DU Mechanical, Thermal and Power Requirements

Parameter	Requirement	Description	Priority
Weight	The gross weight of the gNB-DU must be smaller than 3kg.	Weight requirement	High
Dimension	The dimension of the gNB-DU must be smaller than 3L.	Measurement in dimension	High
Stability	The failure rate of gNB-DU must not exceed 2%	Stability requirement	High
Power Consumption	At full load, the power consumption must not exceed 40W for 2T2R	Power requirement	High
	At full load, the power consumption must not exceed 50W for 4T4R		High
Level of protection	The protection level of gNB-DU is equivalent to the IP31 standard.	Protection level	High
Temperature and moisture	The gNB-DU must be operated and stored under those conditions: Temperature : -5°C~ +55°C moisture : 5%~95%	Ambient temperature and moisture	High
Atmospheric pressure	The gNB-DU must operate normally under the atmospheric pressure between 70 to 106Kpa.	Operational atmospheric	High

		pressure requirement	
Cooling mode	Natural heat dissipation.	Cooling method	High

4.9 AIO White Box Hardware Requirements

This chapter provides the requirements of performance, interface, environmental, EMC, mechanical, thermal and power for AIO white box used within the Indoor Picocell base station. The reference design based on specification shall meet all these requirements based on their priority as specified by operators.

4.9.1 AIO Performance

The performance requirements of the AIO whitebox are listed in Table 4-28.

Table 4-28: AIO Whitebox Performance Requirements

Parameter	Requirement	Description	Priority
Synchronization	Support 1588v2 Synchronization	Timing synchronization method	Low
	Support GNSS Synchronization; Synchronization Between BS $\leq 1.5\mu s$; Carrier frequency error within one subframe $\leq \pm 0.1$ PPM		High
	Support BeiDou Synchronization		High
	Sniffer Synchronization		High
Peak Data Rate	DL: up to 750Mbps UL: up to 340Mbps	Highest theoretical data rate in error free conditions.	High
Channel Bandwidth	≤ 300 MHz OBW	RF Bandwidth	High
Antenna Configuration (Number of Transceivers)	2T2R	Tx/Rx antenna number	High
Conducted Power	up to 27dBm per port	RF power	High

Modulation	DL: QPSK,16QAM,64QAM,256QAM UL: $\pi/2$ -BPSK, QPSK, 16QAM, 64QAM, 256QAM	Modulation schemes	High
Operating band	n41/n78/n79	Radio frequency band	High
Output Power Accuracy	The Output power accuracy shall satisfy the Category B limit defined by the section 6.2.2 in 3GPP TS 38.104 [3].	Power accuracy	High
Tx off Power Level	less than -85dBm/MHz; See section 6.4.1.2 in 3GPP TS 38.104 [3].	Tx power level during an off period	High
EVM at maximum output power	64QAM: EVM smaller than 8% 256QAM: EVM smaller than 3.5%	Max output power	High
Operating band unwanted emissions	The Operating band unwanted emissions shall satisfy the limit defined by the section 6.6.4.2.4 in 3GPP TS 38.104 [3].	RF operation band unwanted emissions requirement	High
Transmitter spurious emissions	The Operating band unwanted emissions shall satisfy the Category B limit defined by the section 6.6.5.2.1 and section 6.6.5.2.4 in [3].	Deliberately generated RF signal by transmitter	High
Receiver Sensitivity	The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel of G-FR1-A1-5; the reference sensitivity levels shall be better than -87.6dBm.	The weakest signal the receiver can identify and process	High
Blocking	In Channel selection, ACS, In-band blocking, out-band blocking, IMD and other receiver specification shall follow the 3GPP guidelines in [3], the reference sensitivity is allowed to degrade at most 6dB under all kinds of interference signal and corresponding level.	Channel selection related requirement	High
Other specifications	Except for all the RF specifications listed above, other RF specifications shall follow the requirement in [3].	Additional standard to comply	High

1 4.9.2 AIO Interfaces

2 The interface requirements of the AIO whitebox are listed in Table 4-29.

3 **Table 4-29: AIO Whitebox Interface Requirements**

Parameter	Requirement	Description	Priority
Transport Interface	1 GbE NG interface to connect with 5G core or 10 GbE NG interface to connect with 5G core	AIO whitebox transport links to 5GC	High

4.9.3 AIO Environmental and EMC

Table 4-30: AIO EMC Requirements

Parameter	Requirement	Description	Priority
EMC	Shall comply with the requirements of 3GPP TS 38.113 (2017-12 Rel-15) [7] for equipment used in telecommunication room	Electromagnetic Compatibility requirement	High
Mounting method	Ceiling, wall, desktop	Mounting requirement	High
Grounding	The AIO whitebox shall support Joint Grounding Method and shall work normally when the grounding resistor is less than 10Ω.	Grounding requirement	High
Environment	The solution should comply with the requirements of O-RAN SuFG Circular economy guidelines on network equipment O-RAN.SuFG.CE [8]	Sustainability requirement	High

4.9.4 AIO Mechanical, Thermal and Power

The mechanical, thermal and power requirements of the AIO whitebox are listed in the following tables.

Table 4-31: AIO Mechanical Requirements

Parameter	Requirement	Description	Priority
Dimensions	≤2.5L	Measurement in three dimensions	Low
Weight	≤2kg	Weight requirement	Low
Status Indicator LED	Shall include as a minimum the following status indicators: ---1 indicating on/off status of the power supply ---1 indicating on/off status of the transmission link	Indicator light	High
Reliability	The MTBF shall be >150000 hours	Mean Time Between Failures	High

Table 4-32: Thermal Requirements

Parameter	Requirement	Description	Priority
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Temperature and Humidity	The solution should support temperature range (-5°C ~ 45°C) and humidity (15%-85%) if implemented indoors.	Environmental requirements for reliability	High
Atmospheric pressure	The whitebox shall operate normally under atmospheric pressure between 70 to 106Kpa.	Operational atmospheric pressure requirement	High
Cooling mode	Passive cooling	System cooling requirement	High
Ingress protection	IP30	Environmental requirements for protection	High

Table 4-33: AIO Whitebox Power Requirements

Parameter	Requirement	Description	Priority
Power Supply	DC 12V with AC adaptor	Power supply Requirement	High
	PoE++	Power supply Requirement	High
Power consumption	Up to 30W	Power consumption Requirement	High